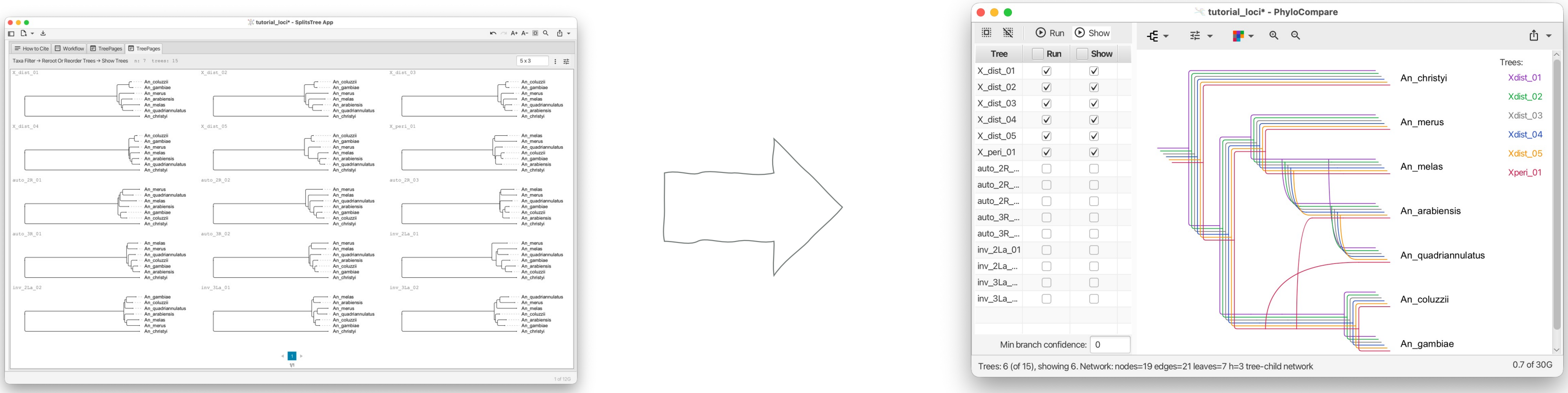
From Trees to Networks



ISMB 2026 Tutorial IP3

Daniel H. Huson, Anupam Gautam and Banu Cetinkaya

Version May 28, 2026

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Your presenters today

Daniel Huson

- Professor of Algorithms in Bioinformatics
- University of Tübingen
- Phylogenetic networks, SplitsTree, PhyloSketch, PhyloCompare

Anupam Gautam

- Post-Doc
- University of Tübingen
- Max-Planck Institute for Biology Tübingen
- PhyloGuide, AI-assisted phylogenetic analysis

Banu Cetinkaya

- PhD student
- University of Tübingen
- Phylogenetic networks, co-developer of PhyloCompare

What we'll do today (09:00 to 13:00)

09:00 - 10:50 Trees

- Per-locus inference with IQ-TREE and BEAST X
- Coalescent species tree with ASTRAL

➢ "We have many gene trees and they disagree"

10:50 - 11:10 / Break ☕

11:10 - 13:00 Networks

- SplitsTree & Neighbor-Net on alignments
- PhyloSketch for interactive network construction
- PhyloCompare for networks from gene tree sets

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Agenda

1. Tutorial overview and dataset
2. Setup verification
3. **Multi-locus phylogenomics lecture**
4. IQ-TREE 3 hands-on
5. BEAST X + SplitsTree posterior visualization
6. ASTRAL hands-on

☕

7. **Networks lecture**
8. SplitsTree Neighbor-Net hands-on
9. PhyloSketch hands-on
10. PhyloCompare hands-on
11. Biological interpretation + wrap-up

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Dataset: the Anopheles gambiae complex

RESEARCH ARTICLE

2 JANUARY 2015

MOSQUITO GENOMICS

Extensive introgression in a malaria vector species complex revealed by phylogenomics

Michael C. Fontaine,[1,2*†] James B. Pease,[3*] Aaron Steele,[4] Robert M. Waterhouse,[5,6,7,8] Daniel E. Neafsey,[6] Igor V. Sharakhov,[9,10] Xiaofang Jiang,[10] Andrew B. Hall,[10] Flaminia Catteruccia,[11,12] Evdoxia Kakani,[11,12] Sara N. Mitchell,[11] Yi-Chieh Wu,[5] Hilary A. Smith,[1,2] R. Rebecca Love,[1,2] Mara K. Lawniczak,[13‡] Michel A. Slotman,[14] Scott J. Emrich,[2,4] Matthew W. Hahn,[3,15§] Nora J. Besansky[1,2§]

Science **347** (6217), 1258524. DOI: 10.1126/science.1258524

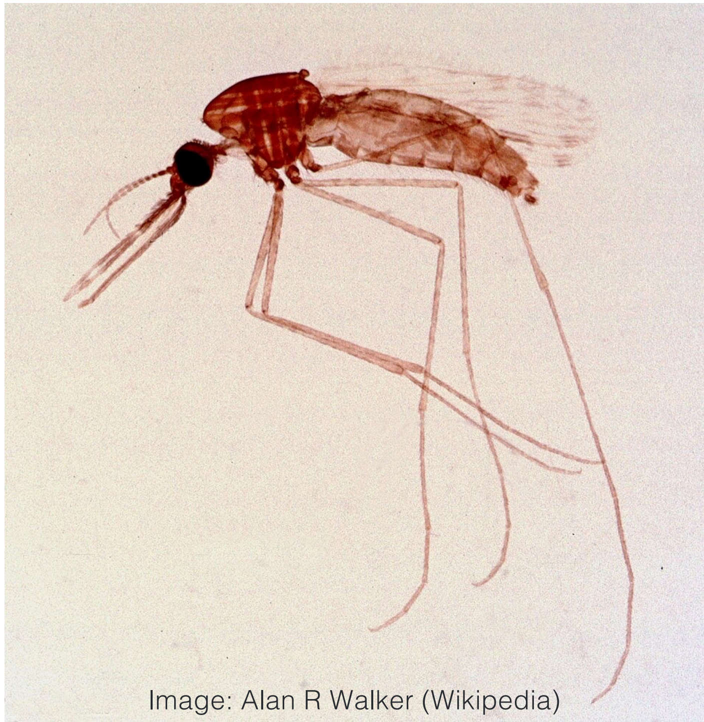
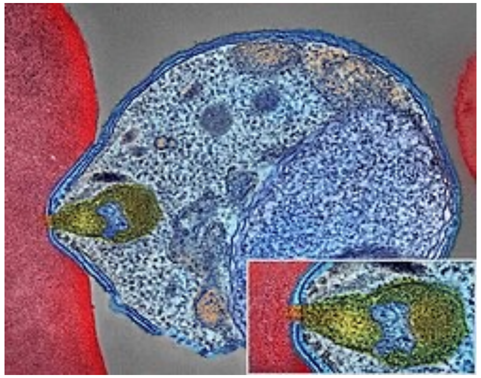


Image: Alan R Walker (Wikipedia)

Dataset: the Anopheles gambiae complex

Why public health cares:

- Three of six species are the principal vectors of *P. falciparum* malaria
- How did they acquire vectorial capacity?
- Can phylogeny help understand the disease



Malaria parasite connecting to a red blood cell
NIAID, Wikipedia

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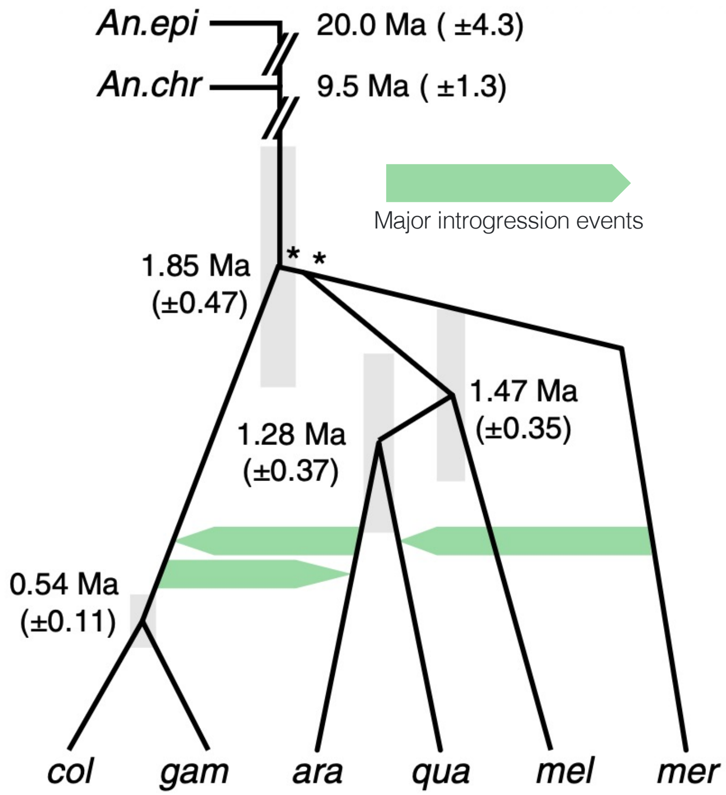
Dataset: the Anopheles gambiae complex

Why phylogenetics cares:

- Six closely related species, recently diverged (about 2 Mya)
- Pervasive autosomal introgression
- X chromosome tells a different story than autosomes
- Several different discordance signals



Fontaine et al (Fig 1C, 2015)

What is introgression?

- Different mosquito species can occasionally hybridize (mate and produce offspring).
- If the hybrid offspring repeatedly mate back into one parental species, then some DNA from the other species can become permanently incorporated.

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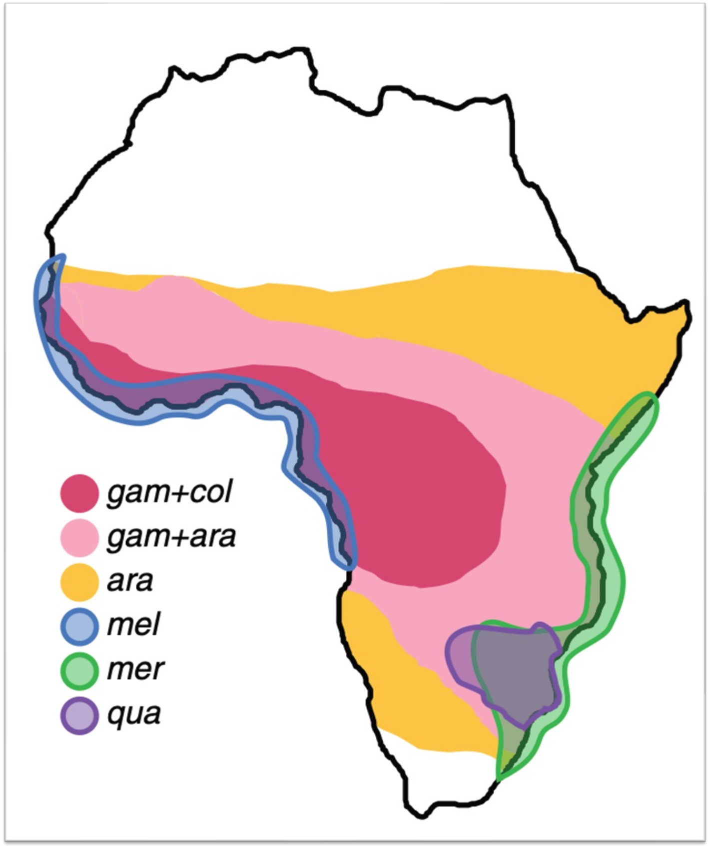
Six species, one complex

- Anopheles gambiae (vector)
- An. coluzzii (vector)
- An. arabiensis (vector)
- An. quadriannulatus (non-vector)
- An. melas (minor vector, brackish water)
- An. merus (minor vector, brackish water)
- An. christyi (outgroup)



Fontaine et al (Fig 1A, 2015)

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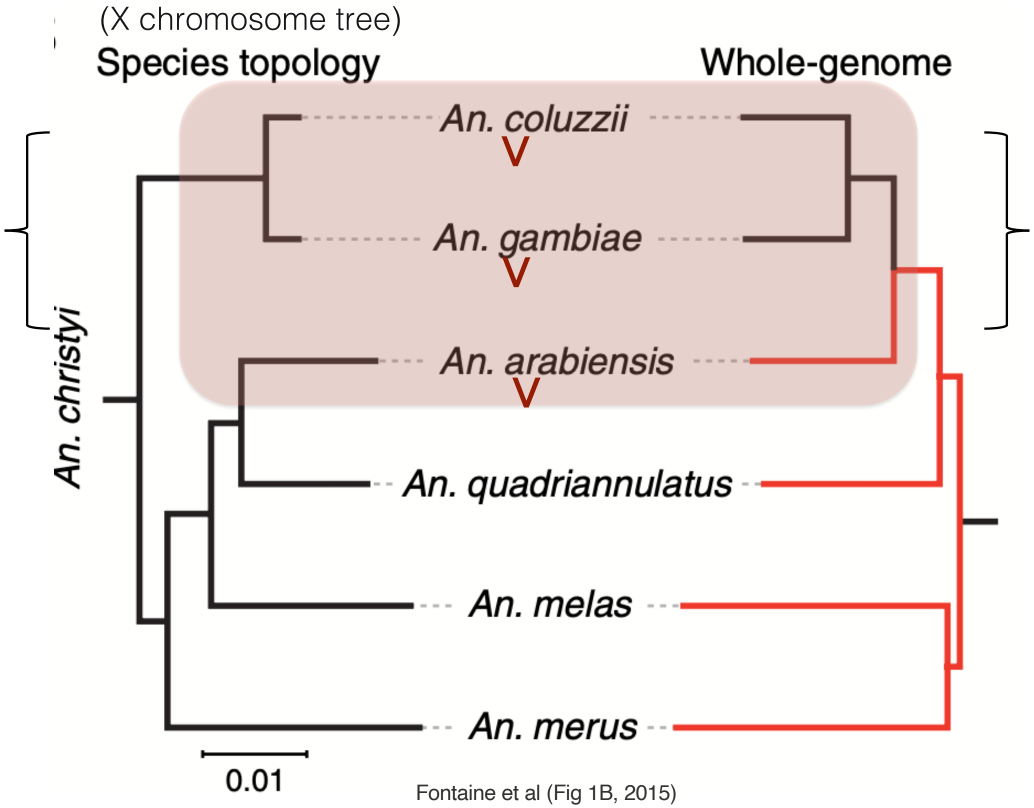
The same dataset, two different trees

Vectors do not form a clade



Vectors form a clade

Fontaine et al (Fig 1B, 2015)

- Both trees 100% bootstrap support, which is “correct”?

The X chromosome tells the truth

Key findings (Fontaine et al, 2015):

1. The X chromosome tree is the species tree.
   Higher sequence divergence between sister pairs confirms this is the historical branching order.
2. The autosomes have been pervasively introgressed between *An. arabiensis* and (*An. gambiae* + *An. coluzzii*).
3. Vectorial capacity may have spread through introgression, not only through de novo mutation.

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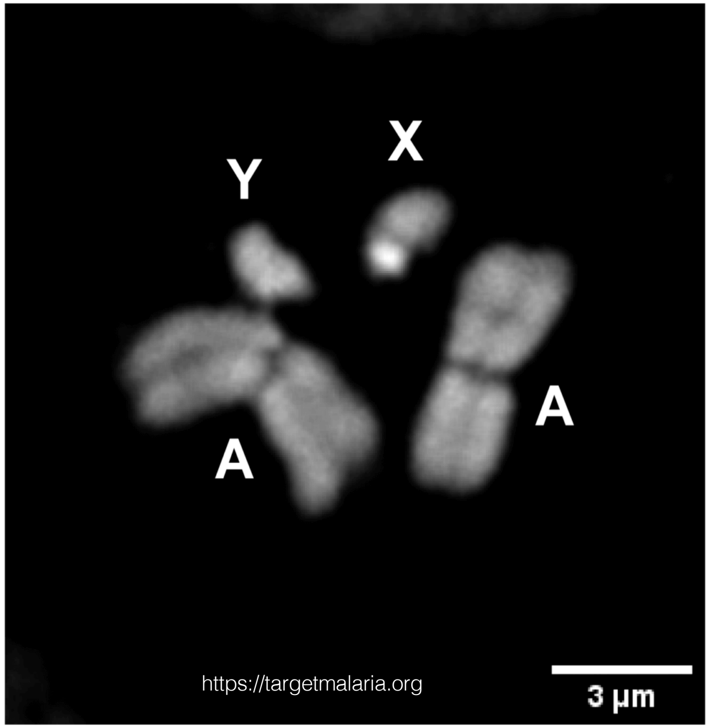
Our data: 15 loci sampled to show the story



Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
Autosomal (2R, 3R)	5	(arabiensis, (gambiae, coluzzii)) - introgressed
2La inversion (2L)	2	2La-specific topologies
3La inversion (3L)	2	(merus, quadriannulatus) - unexpected introgression

Names such as 2La or 3La refer to specific chromosomal inversions

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Software for trees and networks

Part I (trees)

IQ-TREE 3
per-locus ML

BEAST X
Bayesian inference
- BEAUti
- Tracer,
- TreeAnnotator

ASTRAL
MSC species tree

Part II (networks)

SplitsTree
Consensus networks,
Neighbor-Net

PhyloSketch
interactive sketching

PhyloCompare
gene-trees-to-network

Throughout

- PhyloGuide
AI assistant

What we will do today

When the genome has been pervasively introgressed, tree methods can return confident but wrong answers. Networks can make the conflict visible.

- In Part I, we will infer gene trees and a species tree from a multi-locus alignment.
  - We will see that the trees disagree systematically across the genome.
- In Part II, we will use phylogenetic networks to visualize and interpret that disagreement.

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Quick setup verification

☐ Repository cloned:

`git clone https://github.com/husonlab/trees-to-networks-tutorial.git`

☐ `iqtree3 --version` works

☐ `java -jar /path/to/astral.5.7.8.jar -h` works

☐ `beauti` launches

☐ SplitsTree launches

☐ `tracer` launches

☐ PhyloSketch launches

☐ PhyloCompare launches

☐ PhyloGuide URL bookmarked

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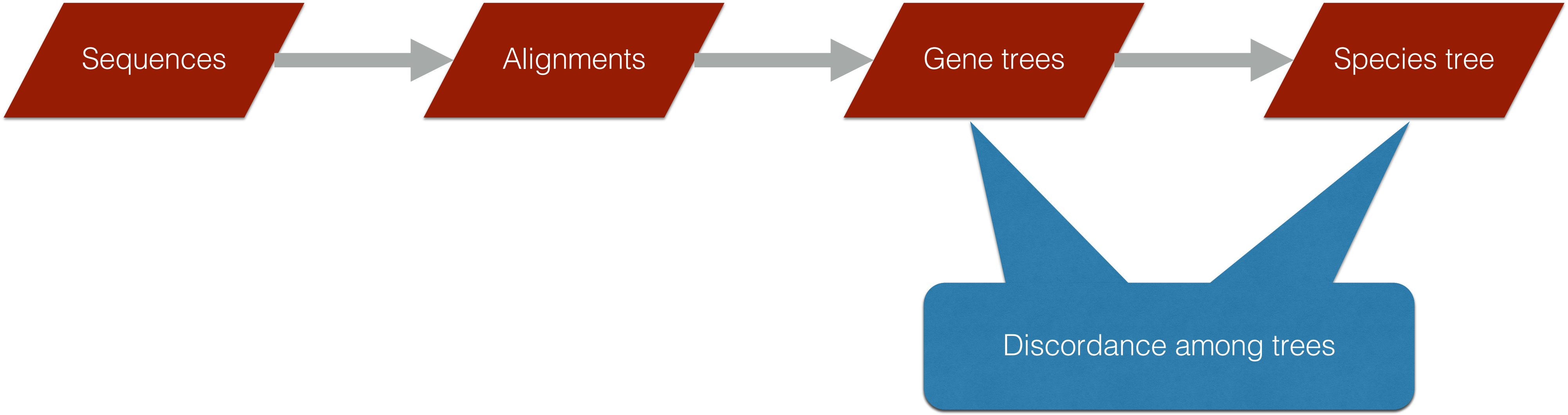
From sequences to gene trees to species trees



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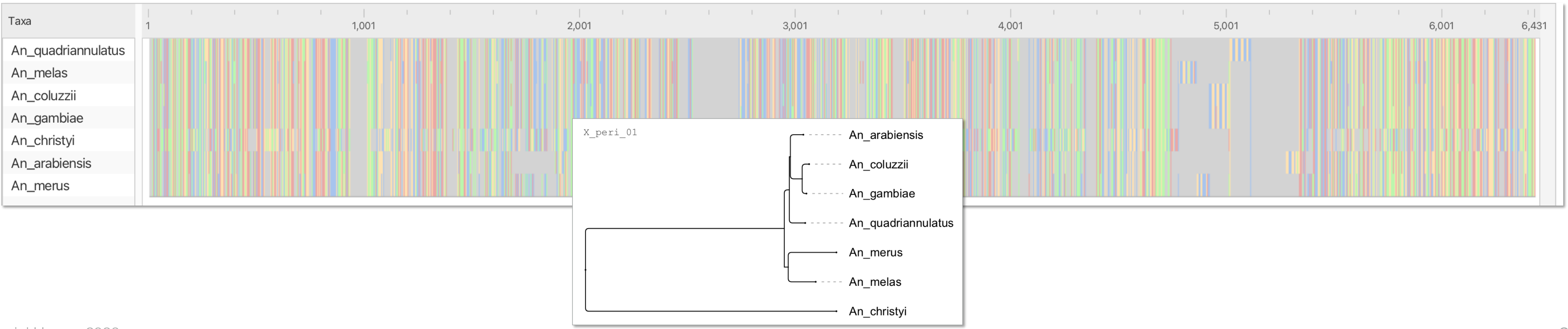
At a single locus

- A multiple sequence alignment (MSA) is our raw input
- 7 species, ~5000 nucleotide positions, every column is informative about how the species relate
- From the MSA we infer a **gene tree**: the evolutionary history of this one stretch of DNA



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Inferring a tree by maximum likelihood

- Likelihood: probability of observing the alignment A given tree T and substitution model M:

$$P(A \mid T, M)$$

- Find the tree and model that maximize this probability
- Tool: IQ-TREE 3
- Substitution models (JC, K2P, HKY, GTR, +G, +I)
  how nucleotides change over time
- Bootstrap: assess confidence in the tree (we use UFBoot)

Inferring a *distribution* of trees by Bayesian inference

- Bayesian inference returns the full posterior distribution
- Not just one best tree but P($T,M \mid A$) for "every" possible tree
- Tool: BEAST X (uses MCMC sampling)

➢ Thousands of trees, each weighted by posterior probability

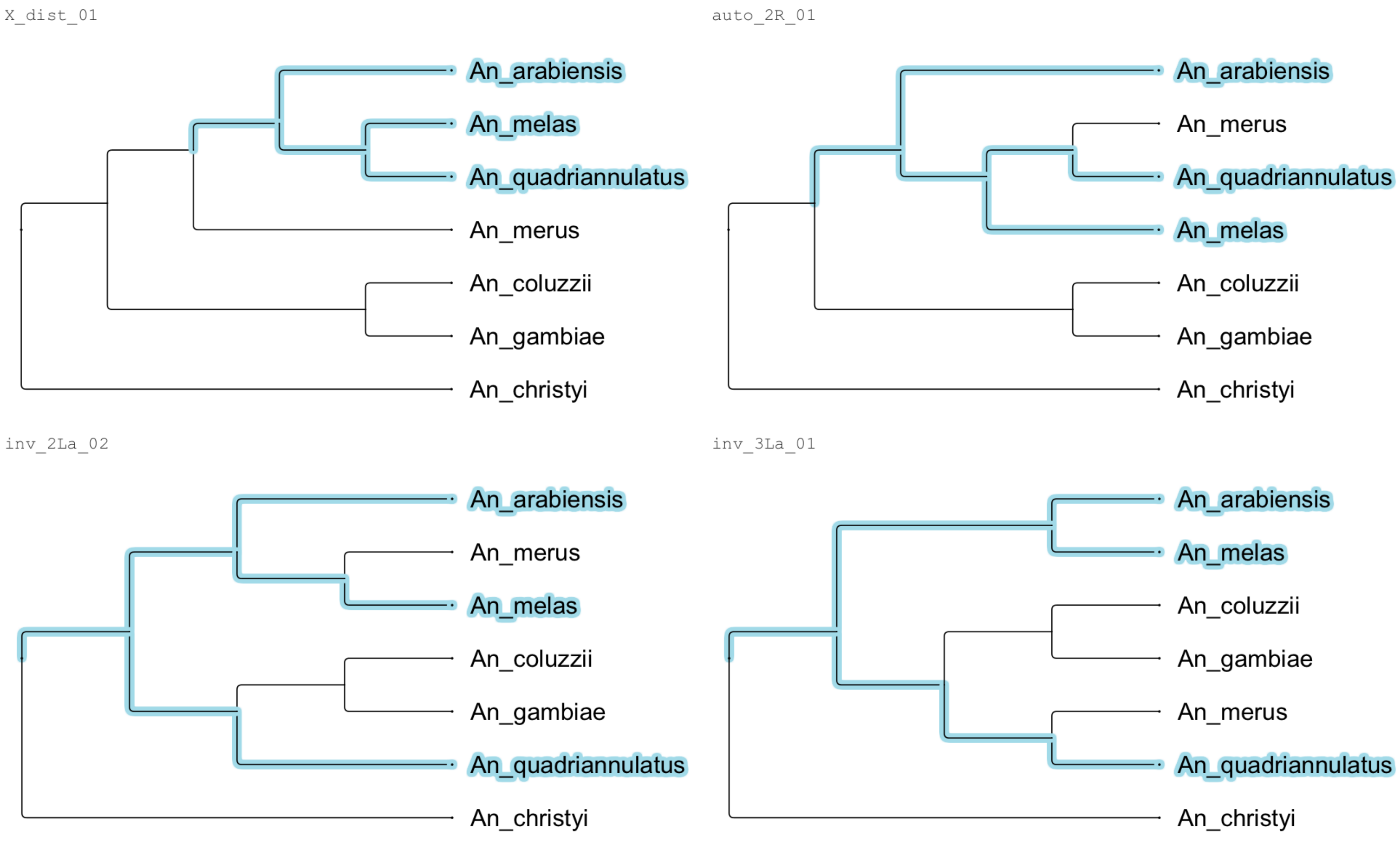
- Captures uncertainty natively

Genome-scale data: many loci, many gene trees

Phylogenomics:

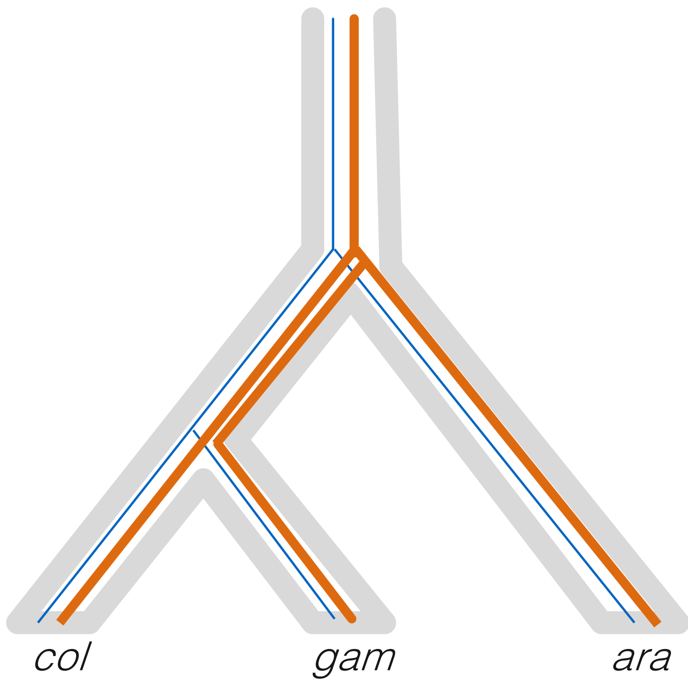
- hundreds to thousands of loci per analysis
- Each locus has its own evolutionary history
- Run ML or Bayesian inference per locus -> a *set* of gene trees
- For our tutorial: 15 loci, 15 gene trees

Gene trees disagree



Cause #1 of discordance: incomplete lineage sorting (ILS)

- Ancestral polymorphism can persist across multiple species
- Different alleles get inherited by different descendant lineages
- The gene tree can have a different topology than the species tree
- Purely stochastic; no gene flow required



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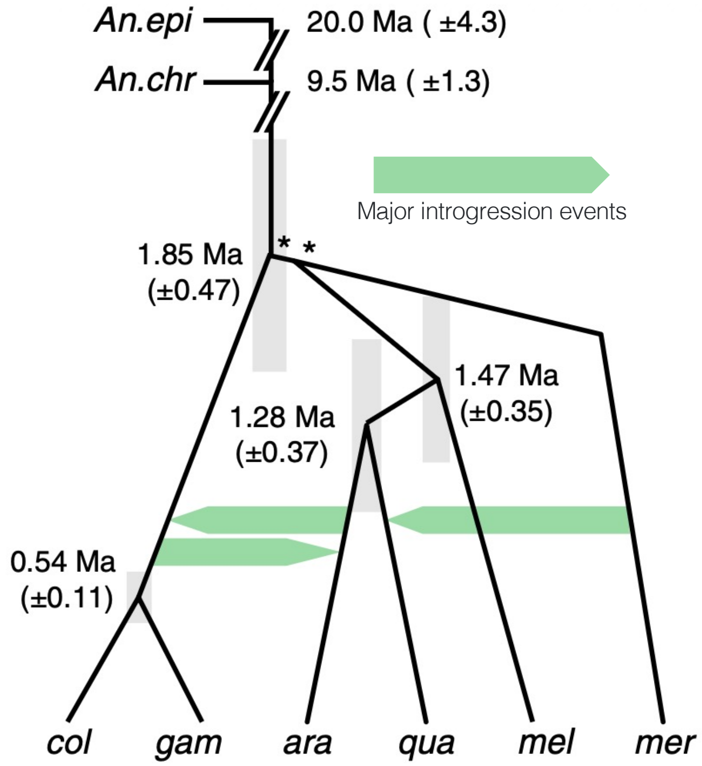
Cause #2 of discordance: introgression

- After speciation, gene flow between species can re-introduce alleles
- Hybridization, horizontal transfer
- Introgression is a biological process not just stochastic
- Introgressed alleles look like the donor species, not like closest relatives
- Often non-uniform across the genome (only some regions introgressed)



Fontaine et al (Fig 1C, 2015)

Telling ILS and introgression apart

- Both ILS and introgression produce gene trees that disagree with the species tree
- ILS: the alleles that don't sort are ancestral, so they coalesce *deep* in the species tree
- Introgression: the alleles came from a sister species recently, so they coalesce *shallow*
- This is how Fontaine et al. (2015) showed Anopheles autosomes are introgressed, not just ILS

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Now: let's try to see this in our data

- 15 loci spanning autosomes, X chromosome, and inversion regions
- Run IQ-TREE on each -> 15 gene trees
- We expect to see discordance
- Question: where does the discordance occur?

Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
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PhyloGuide Q1

PhyloGuide
PHYLOGENETICS, GROUNDED

Q: We have multi-locus alignments from a recently diverged species complex. How should we infer a species phylogeny?

A: Infer gene trees for each locus, then estimate a species tree with ASTRAL instead of relying only on concatenation. This accounts for gene-tree discordance caused by ILS.

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Hands-on: per-locus ML inference with IQ-TREE 3

- Time budget: ~20 minutes
- Goal: produce 15 maximum-likelihood gene trees, one per locus, and see how they vary across the genome
- Where we are: `cd trees-to-networks-tutorial`
- What we'll run: `iqtree3 -S data/alignments/ ...`

  (full command on next slide)

The IQ-TREE 3 command

```
iqtree3 -S data/alignments -B 1000 --prefix tutorial_loci -T AUTO -st DNA
```

Flag	What it does
`-S data/alignments`	Per-locus mode: one tree per FASTA in this directory
`-B 1000`	Ultrafast bootstrap with 1000 replicates
`--prefix tutorial_loci`	Output filename prefix
`-T AUTO`	Auto-pick number of threads
`-st DNA`	Force DNA mode (auto-detection fails on some loci)

Run the IQ-TREE 3 command

Run the command.

- What runs internally:
  - ModelFinder picks the best substitution model per locus
  - then maximum-likelihood tree search
  - then UFBoot for branch support
- Runtime: ~ 5 seconds
- Output file: `tutorial_loci.treefile`

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Run the IQ-TREE 3 command

- You'll see one block of output per locus, ending like this:

```
Subset  Type   Seqs   Sites  Infor  Invar  Model  Name
1    DNA     7   6208    144     5068       X_dist_01.fasta
2    DNA     7   7530    234     5899       X_dist_02.fasta
...

15    DNA  7    5454 105  4455      inv_3La_02.fasta
```

- What's normal:
  - All 15 loci have 7 sequences
  - Most loci pick HKY+F+G4 as best-fit model
  - Christyi (the outgroup) failing composition chi2 tests in several loci is not a problem

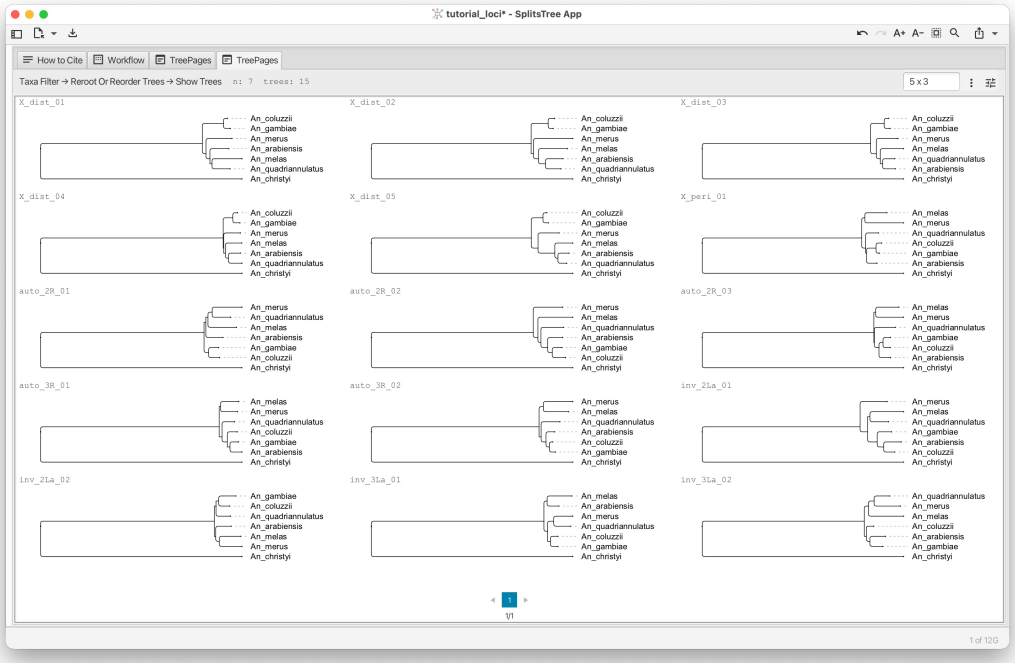
Nexus file for SplitsTree and PhyloCompare

- IQ-TREE wrote 15 Newick trees to `tutorial_loci.treefile`, one per line in alphabetical order of FASTA filenames
- For SplitsTree and PhyloCompare, we want a NEXUS file with each tree labeled by its locus ID and correctly rooted by outgroup An_christyi
- Run our helper:

```
python tools/make_nexus_treeset.py \
    --alignments-dir data/alignments/ \
    --treefile tutorial_loci.treefile \
    --out tutorial_loci.nex\
    --outgroup An_christyi
```

Open the gene trees in SplitsTree and look around

- Open `tutorial_loci.nex` in SplitsTree
  - File -> Open -> select the .nex file
- What you should see: 15 named trees in the tree set



Discussion items

- Do all loci show the same topology? Why not?
- Can you tell apart X-distal from autosomal loci just by topology?
- Where in the genome would you expect to see "species-tree" signal vs "introgressed" signal, based on what we know about this dataset?

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PhyloGuide Q2

PhyloGuide
PHYLOGENETICS, GROUNDED

Q: We ran IQ-TREE on 15 alignments and the gene trees disagree across chromosomes - X chromosome shows one topology, autosomes another. What does this mean?

A: Different genomic regions can have different evolutionary histories due to ILS, introgression, selection, or gene-tree error. It does not mean one topology is necessarily correct.

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Hands-on: Bayesian inference + visualizing the posterior

- Time budget: 30 minutes
- Goal: produce a posterior tree distribution for one locus, then visualize the within-locus uncertainty as a network
- Tools: BEAST X, BEAUti, Tracer, TreeAnnotator, SplitsTree
- Locus: X_dist_04 (one of our cleanest X-distal alignments)

BEAST X
Bayesian Evolutionary Analysis Sampling Trees

From “best tree” to “distribution of trees”

- What ML gave us (IQ-TREE, previous segment):
  - One tree per locus
  - Bootstrap support values
  - Discordance ACROSS loci
- What Bayesian gives us (BEAST X):
  - A posterior distribution of trees per locus
  - Captures uncertainty WITHIN a locus
  - Trees weighted by posterior probability
  - We can summarize as one tree OR visualize the full distribution

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We'll run BEAST X on a single locus: X_dist_04

- Locus: chrX:9,845,592 - 9,849,716 (4,872 bp), X-distal Xag region
- Expected topology: (arabiensis, quadriannulatus) sister, with christyi outgroup
- From IQ-TREE earlier: HKY+F+G4 best-fit model
- Why this locus: clean signal, fast Bayesian convergence, represents the "true species tree" topology
- Why one locus only: a Bayesian run on all 15 loci would take the full tutorial; we use one as a worked example

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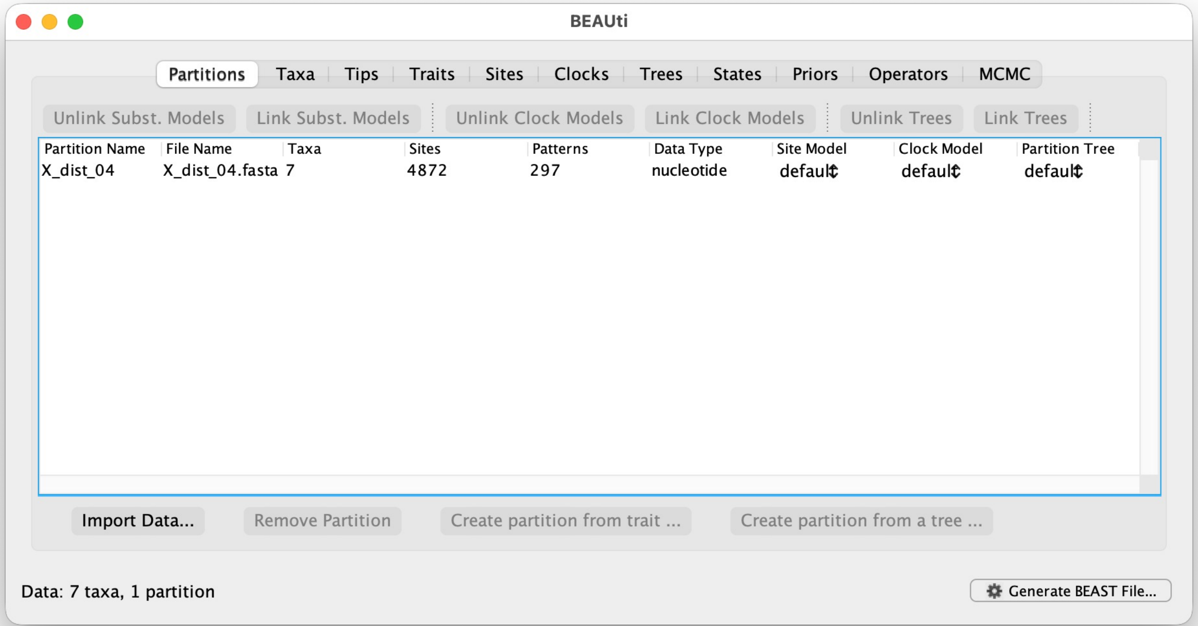
BEAUti: import data and set substitution model

1. **Launch BEAUti** (from BEAST X installation)
2. **File -> Import Data ->** `data/alignments/X_dist_04.fasta`



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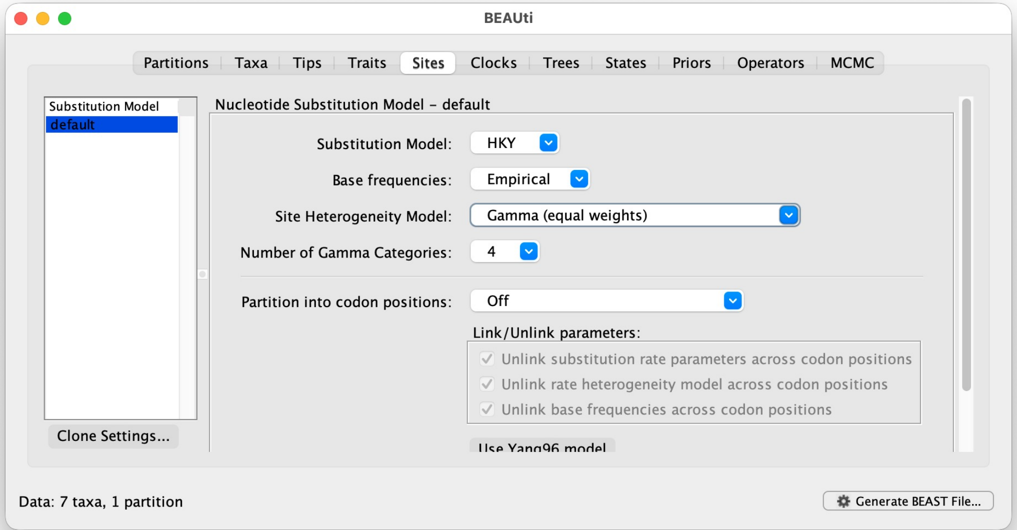
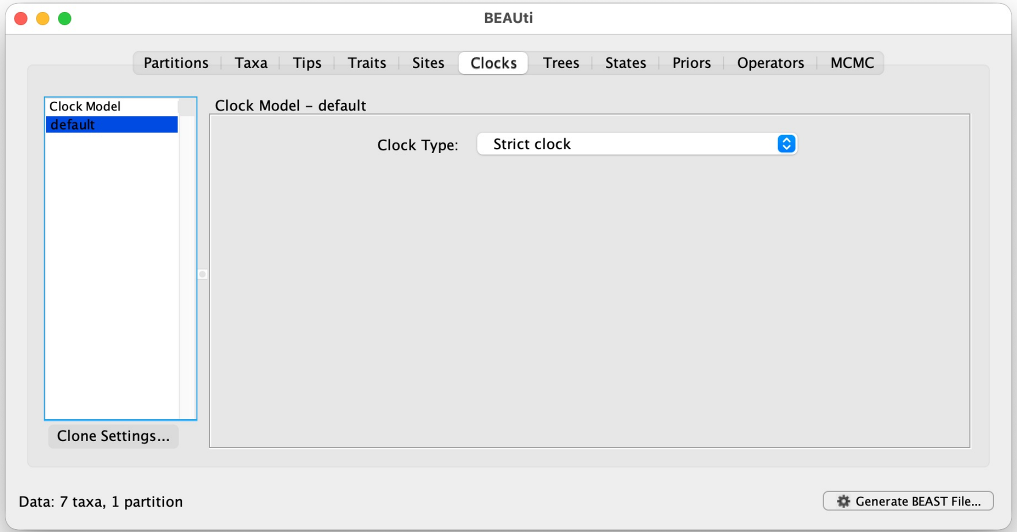
BEAUti: import data and set substitution model

3. **Sites tab**:
   - Substitution Model: **HKY**
   - Base frequencies: **Empirical**
   - Site Heterogeneity Model: **Gamma**
   - Number of Gamma Categories: **4**
   - Partition into codon positions: **off**



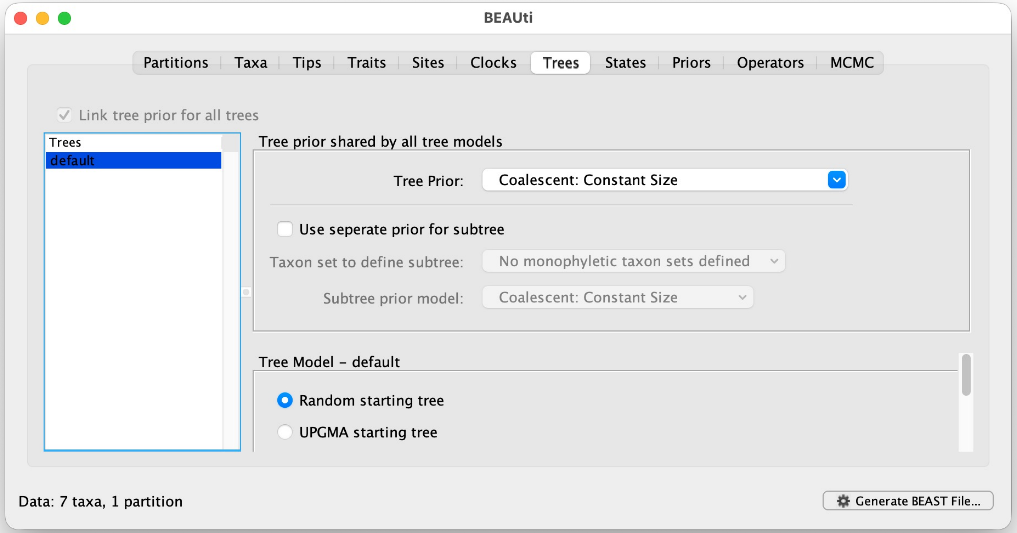
BEAUti: clock model and tree prior

4. Clocks tab:
   - Clock Type: **Strict clock**



BEAUti: clock model and tree prior

5. Trees tab:
   - Tree Prior: **Coalescent: Constant Size**
   - This is appropriate for the within-genus timescale

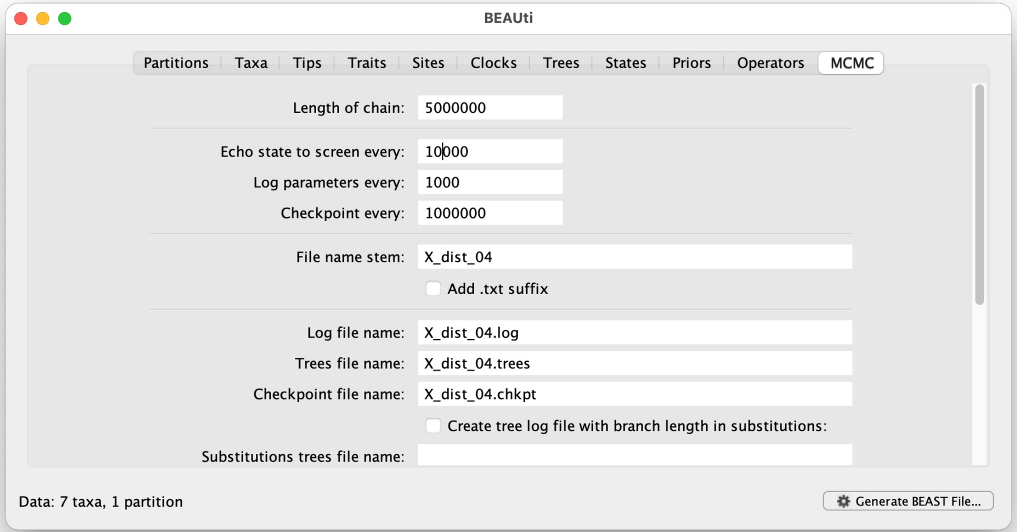


BEAUti: clock model and tree prior

6. **States tab**: nothing to change
7. **Priors tab**: leave at defaults
8. **Operators tab**: leave at defaults

MCMC settings, generate XML, and start BEAST

9. MCMC tab:
   - Length of chain: **5,000,000** (short for live demo; production runs would use 20M)
   - Echo state to screen every: 10,000
   - Log parameters every: **1000** (gives 5,000 samples)
   - File name stem: **X_dist_04**



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MCMC settings, generate XML, and start BEAST

10. File -> Generate BEAST File -> save as `X_dist_04.xml`
11. Run BEAST X immediately in a terminal:

    `beast -beagle -overwrite X_dist_04.xml`

    - Expected runtime: about 2-3 minutes on Apple Silicon with BEAGLE
    - Let it run in the background. We will come back to it later.

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Inspect a precomputed run in Tracer (while yours runs)

- Your live run is producing `X_dist_04.log` (parameter trace) and `X_dist_04.trees` (sampled trees), but it is not done yet
- To see what good convergence looks like NOW, open the precomputed long run from the repo:

```
Tracer

File -> Import Trace File

-> precomputed/beast/X_dist_04.log
```

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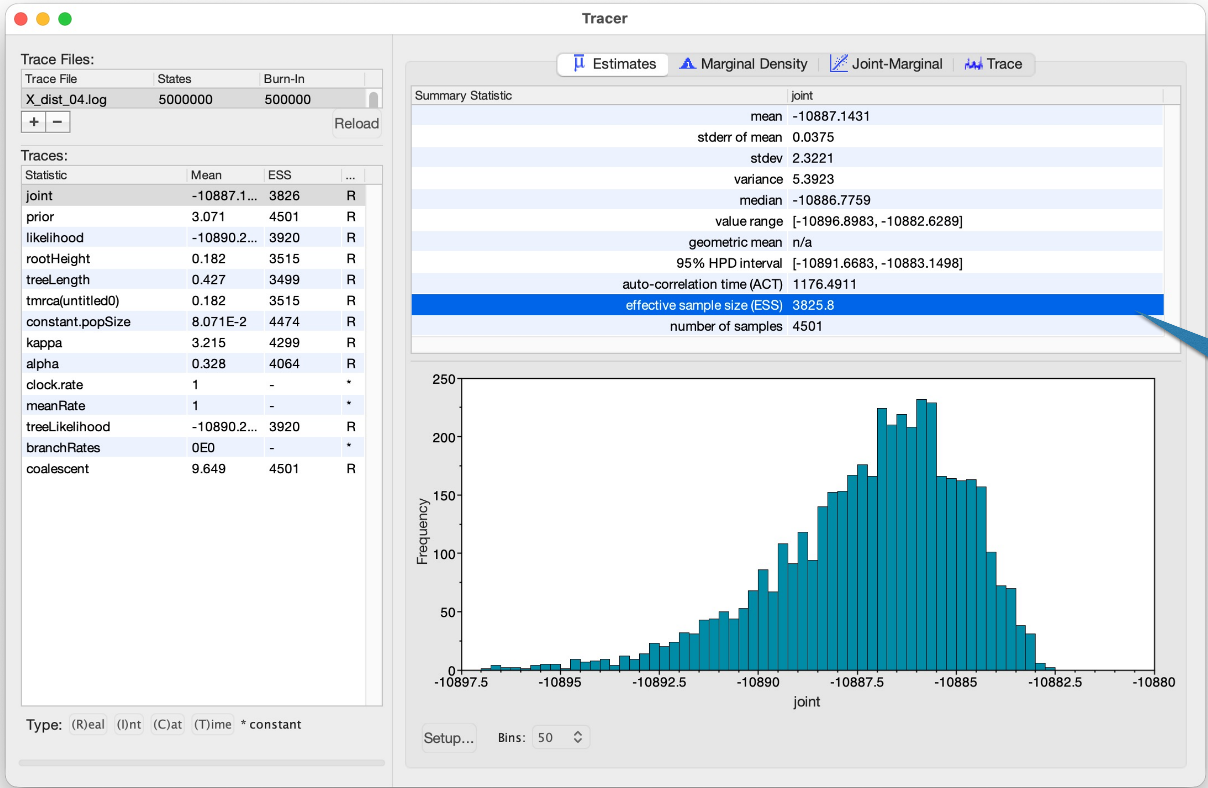
Inspect a precomputed run in Tracer (while yours runs)



Effective sample size is good (nearly 4000)

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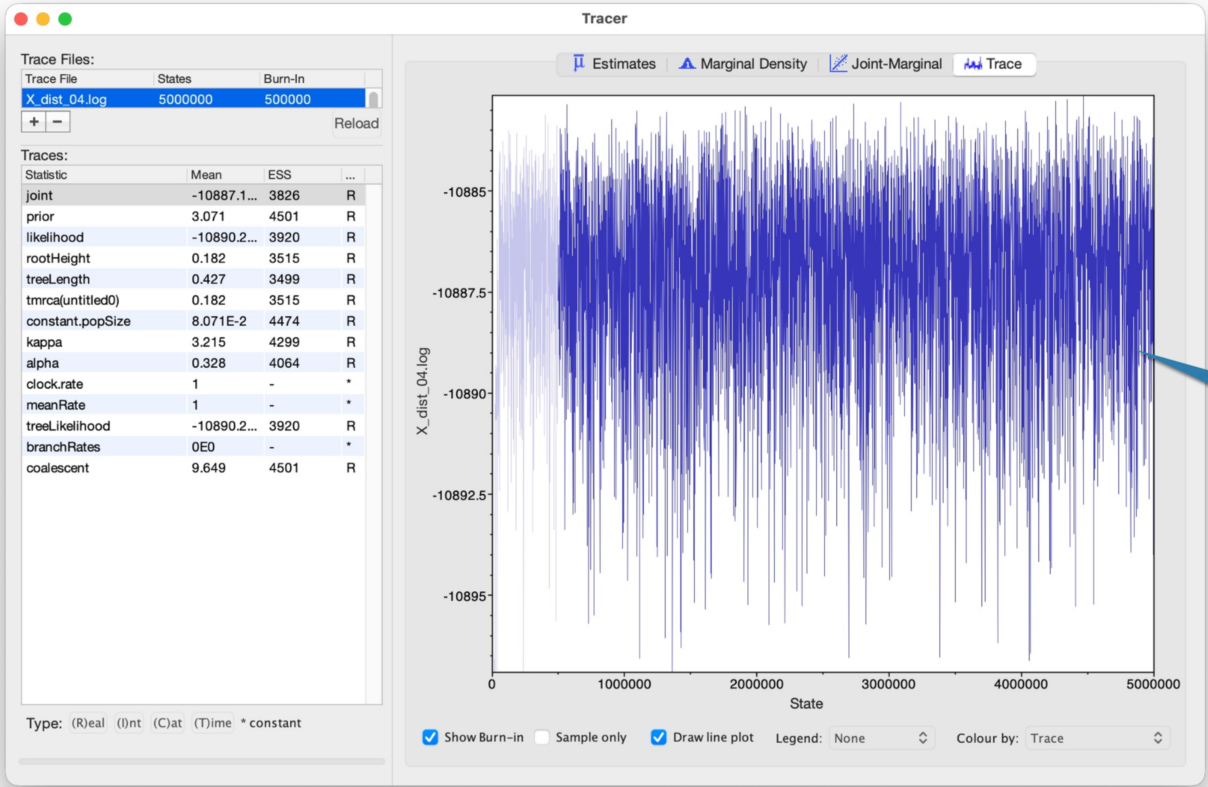
Inspect a precomputed run in Tracer (while yours runs)



Sampling looks good
("hairy caterpillar)

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Summarize the posterior with TreeAnnotator

- We have a distribution of trees but often want one summary tree
- TreeAnnotator computes the Maximum Clade Credibility (MCC) **tree**: the tree from the posterior whose clades collectively have the highest product of posterior support

```
treeannotator -burnin 500 -heights median \
    precomputed/beast/X_dist_04.trees X_dist_04.MCC.tre
```

`-burnin 500`	Discard first 500 trees
`-heights median`	use median node heights for branch lengths
`precomputed/beast/X_dist_04.trees`	Input trees
`X_dist_04.MCC.tre`	one tree with posterior probabilities annotated on each internal node

Summarize the posterior with TreeAnnotator

- By the time you see this slide, your live BEAST run should be done.

- Replace `precomputed/beast/X_dist_04.trees` with `X_dist_04.trees` to use your own run instead.

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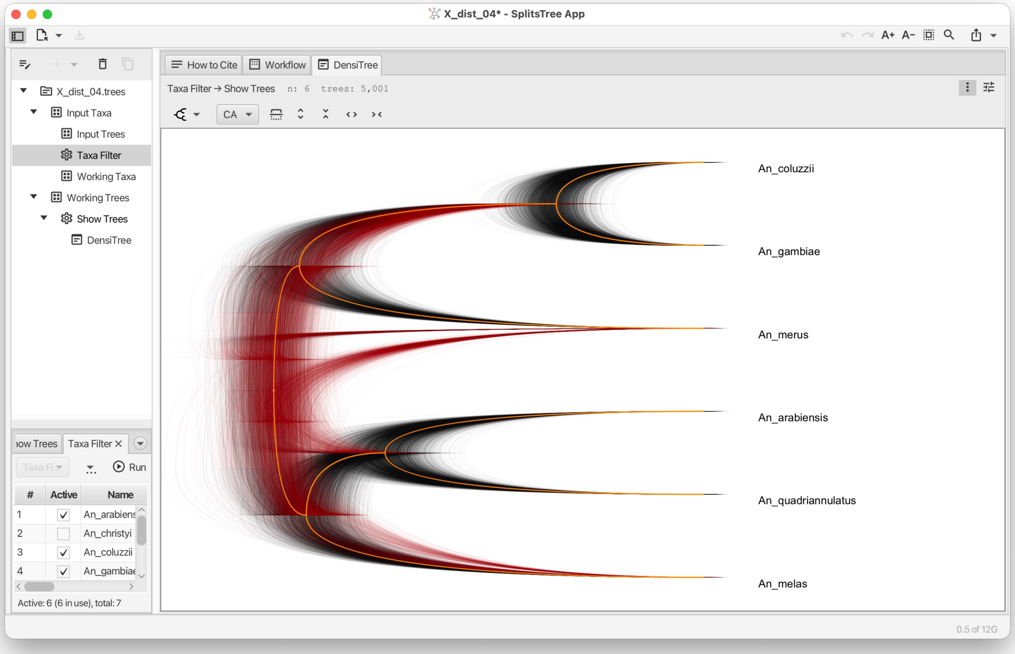
Visualizing the full posterior: DensiTree in SplitsTree

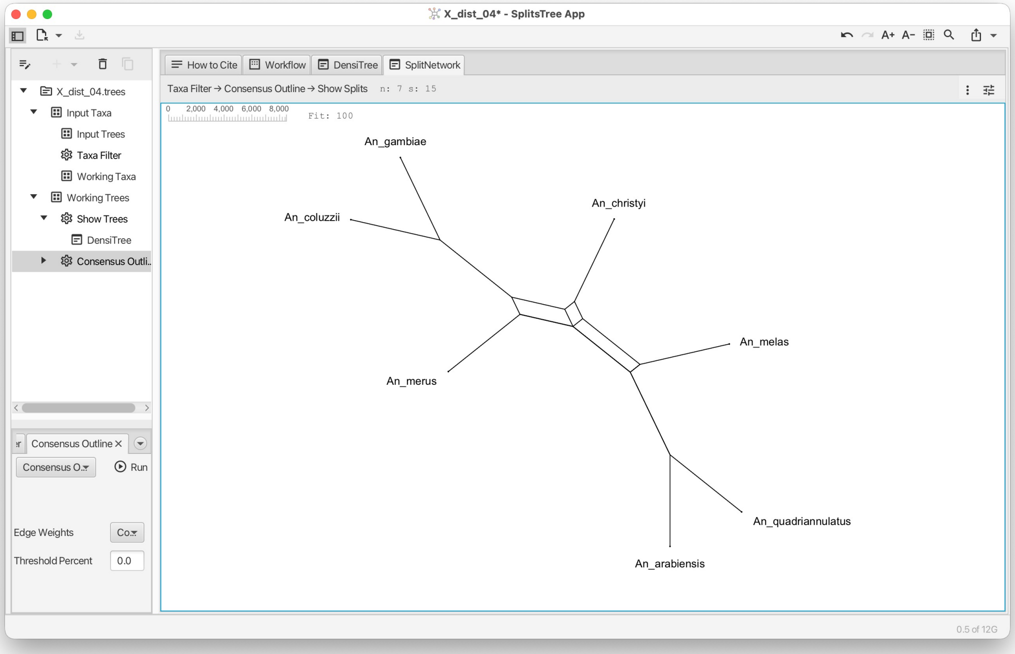
- Open SplitsTree
- File -> Open -> select `X_dist_04.trees` (the full posterior, 5000 trees)
- Tree -> Show DensiTree to overlay all trees



Your first network: consensus outline from the posterior

- In SplitsTree, with the same posterior tree set loaded:
  - Network menu > Consensus Outline

➤ Network with splits weighted by posterior support (counts)



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Hands-on: coalescent species tree with ASTRAL

- ASTRAL: for a set of gene trees, returns a single species tree that maximizes quartet agreement under the multispecies coalescent (it models ILS but NOT introgression)
- Input: `tutorial_loci.treefile` (output of iqtree)
- Command:

```
java -jar astral.5.7.8.jar \
    -i tutorial_loci.treefile \
    -o tutorial_loci.ASTRAL_species_tree.tre
```

➢ Species tree with quartet support values at each internal node

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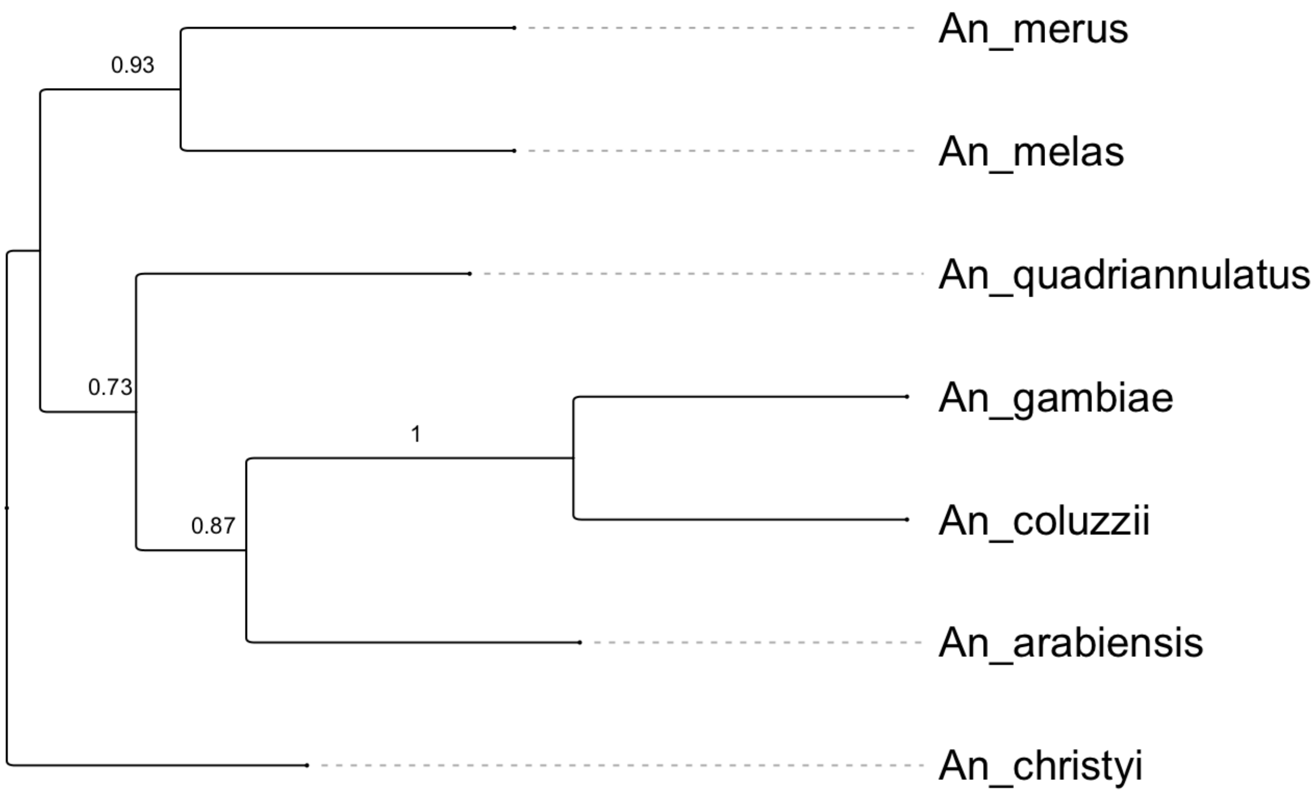
What does ASTRAL tell us?



(Re-rooted on `An_christyi`; ASTRAL output is unrooted, so the Newick file looks different but the topology is the same.)

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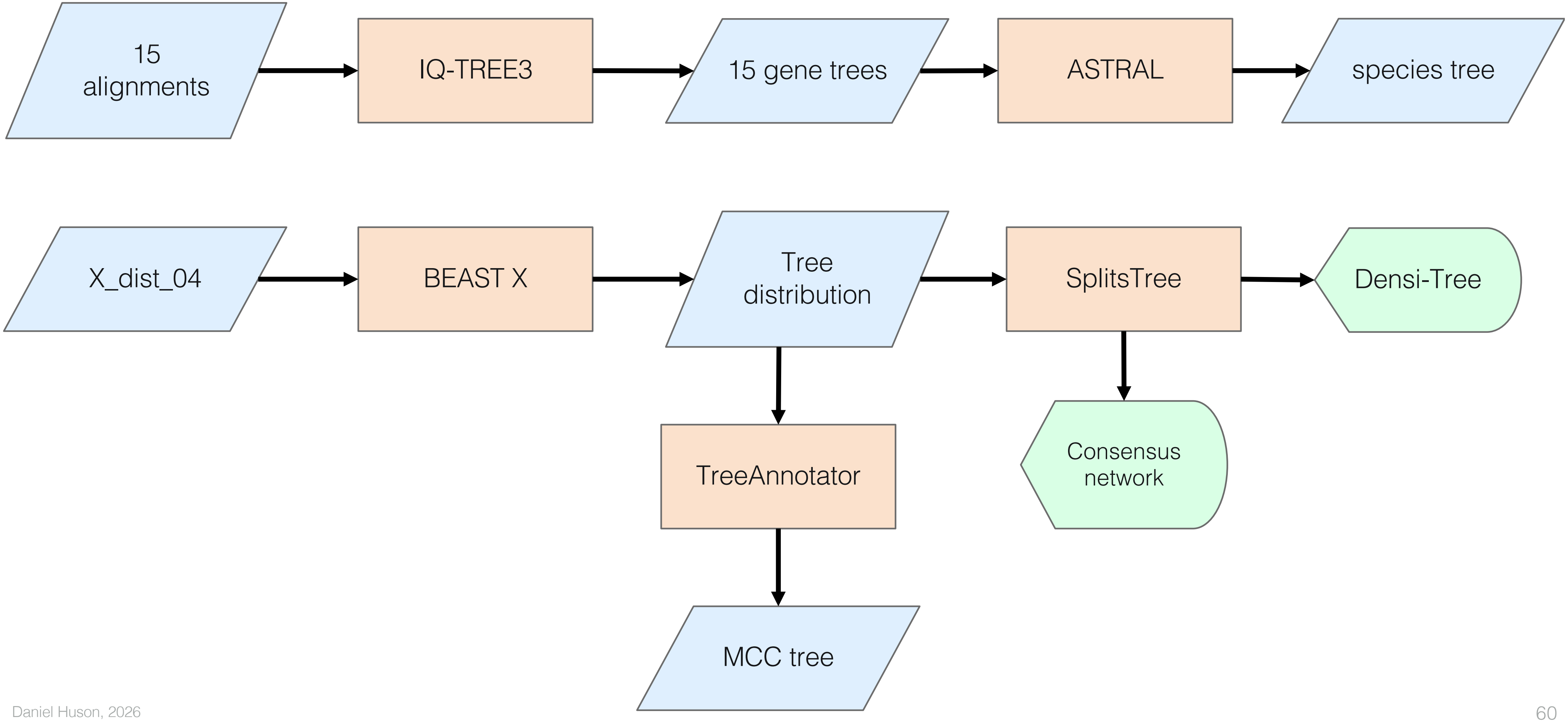
Part I recap: from sequences to trees



But we know one of our gene tree categories is right

- The X-distal gene trees give the `(arabiensis,quadriannulatus)` topology
- We know from Fontaine that THIS is the true species branching order
- The autosomal gene trees and ASTRAL show the introgression-driven topology
- ASTRAL gave us a confident answer that is probably biologically wrong
- ASTRAL is *not broken* - it is doing exactly what its model says it should do
- The data contain *introgression* that the model (MSC, ILS only) can't represent

PhyloGuide Q3

Q: When is a tree inadequate? When do we need a network?

PhyloGuide
PHYLOGENETICS, GROUNDED

A: A tree is inadequate when evolution is not strictly branching, such as with hybridization, introgression, HGT, or strong conflicting signal. Networks capture mixed ancestry and conflicting histories.

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After the break: Making conflict explicit

- **Part II preview**:
  - SplitsTree Neighbor-Net: networks of conflicting splits in the alignment data
  - PhyloSketch: build a hypothesis network interactively
  - PhyloCompare: networks inferred from the gene trees - and SEE the introgression
- **Break**: 20 minutes, then 11:10 reconvene

COFFEE BREAK

20 minutes

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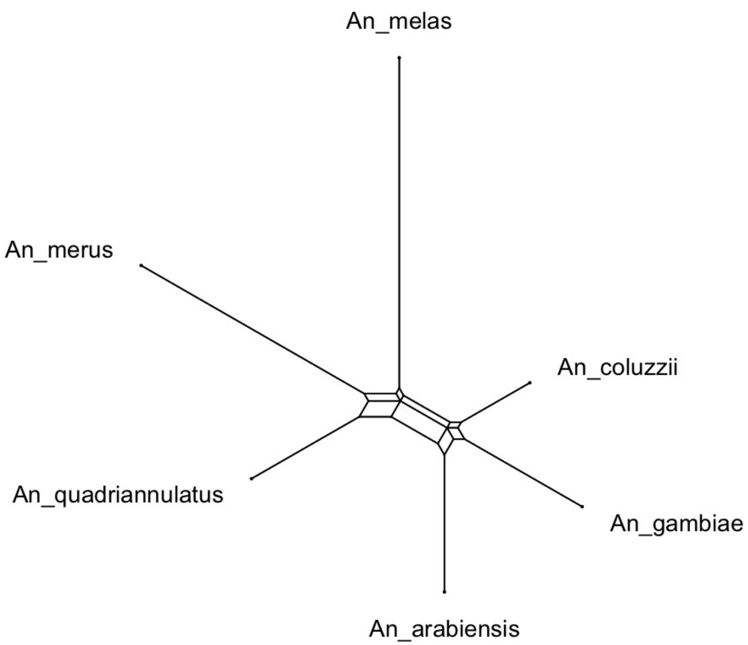
Part II: Networks and cross-locus reticulation

- Where we left off:
  - 15 gene trees, systematically disagreeing
  - ASTRAL gave a confident species tree
  - But it is the introgression-driven topology, not the true one
  - We have evidence the data contain a non-tree-like signal
- What we will do next:
  - Three network views of our data, each with a different tool
  - Networks visualize the conflict explicitly
  - We will *see* the introgression rather than infer it indirectly

Split networks

- A split partitions the taxa into two sets
  - Each edge in a phylogenetic tree corresponds to a split
- A tree has a compatible set of splits
  - no two splits are incompatible
- A split network can represent incompatible splits
  - Any two incompatible splits produce a parallelograms in a split network



Building a split network from a distance matrix

- Input: a pairwise distance matrix (computed from an alignment)
- Neighbor-Net (Bryant & Moulton 2004, extends Neighbor-Joining)
- Instead of forcing each step to produce a tree, it can retain conflicting evidence as parallelograms
- Output: a split network that approximates the input distances
- Tool: SplitsTree

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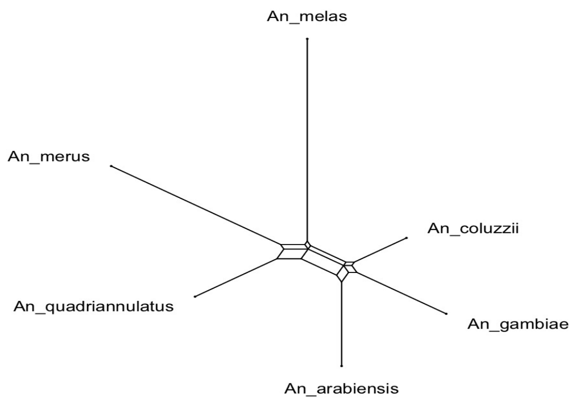
Two roles for networks: implicit and explicit

1. Implicit networks

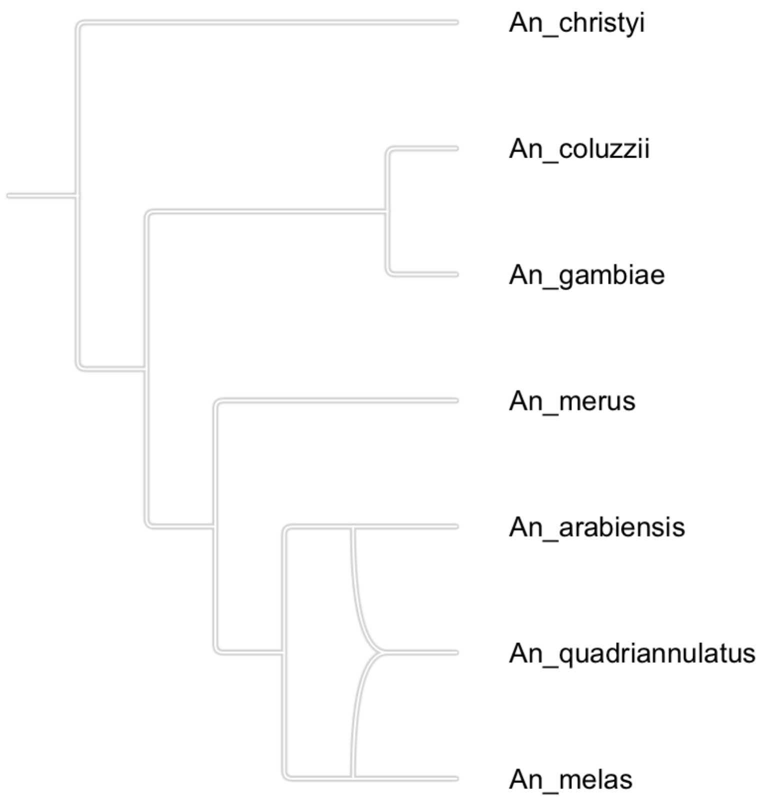
- Visualize conflicting signal without claiming specific evolutionary events
- "Here is conflict" - they don't say what caused it
- Examples: split networks

➢ Exploration, visualization, detecting non-tree-like signal



Rooted phylogenetic networks

- A rooted phylogenetic tree is a tree with
  - One root, leaves are taxa, internal nodes are inferred ancestors
- A phylogenetic network generalizes this:
  - Internal nodes can have more than one parent (reticulation nodes)
  - Can represent both vertical inheritance *and* events where lineages combine or exchange genetic material.



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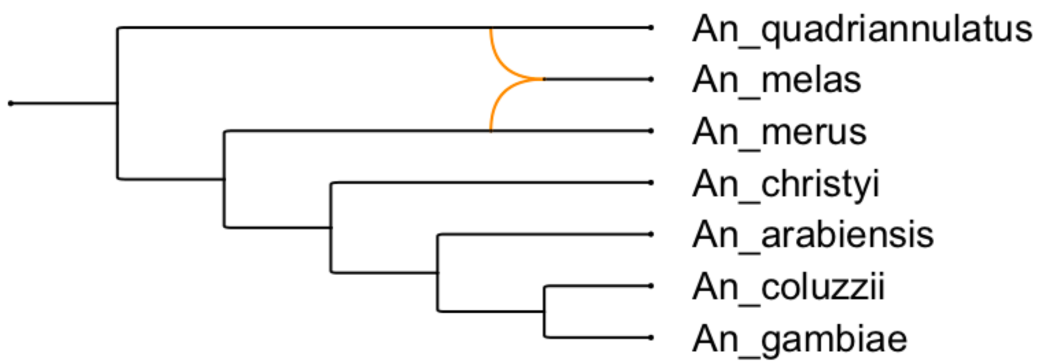
Two roles for networks: implicit and explicit

2. Explicit networks

- Represent specific evolutionary events: hybridization, introgression, horizontal transfer
- Have reticulation nodes corresponding to those events
- "These two parent lineages produced this hybrid"
- Examples: PhyloFusion output, hybridization networks

➢ Infer biological history



Explicit reticulation: where two lineages meet

- A reticulation node in an explicit network represents an event where genetic material from two parent lineages combined into one descendant
- This may correspond to:
  - Hybridization: two species produce a hybrid that becomes a new lineage
  - Introgression: gene flow between two species after their initial divergence
  - Horizontal gene transfer: genetic material moves between non-sister taxa
  - Recombination: blocks of DNA from different parents
- A network can have multiple reticulation nodes - one per event

You have already computed a network

- In Part I, the consensus network from the BEAST X posterior:
  - Each split weighted by posterior probability
  - Parallelograms appear where the posterior is uncertain
  - This was an implicit network of within-locus uncertainty
- Today we will compute several more, all looking at our 15 gene trees from different angles

From gene trees to a reticulate network

- PhyloFusion takes as input a set of rooted gene trees and computes an explicit network
- Finds a network that displays all input trees with as few reticulations as possible
- Each reticulation in the output corresponds to a hypothesized reticulation event that explains why the gene trees disagree
- Interactive: you can select which gene trees to include in the inference, and which to display *inside* the inferred network

(Zhang, Cetinkaya, Huson 2026)

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Our three network views today

1. **SplitsTree Neighbor-Net on alignments** (next, 20 min)
   - Input: alignment data
   - Output: implicit split network of conflicting splits
   - Goal: see the raw conflict signal in the data
2. **PhyloSketch interactive sketching** (30 min in)
   - Input: your biological intuition + the gene trees you have seen
   - Output: a hand-drawn hypothesis network
   - Goal: think like a phylogeneticist building a hypothesis
3. **PhyloCompare on the gene tree set** (60 min in)
   - Input: the 15 gene trees from IQ-TREE
   - Output: explicit reticulation network (using PhyloFusion)
   - Goal: see introgression events inferred directly from the trees

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Hands-on: Neighbor-Net on the 15 alignments

- Time budget: 20 minutes
- Goal: visualize the conflicting signal directly in alignment data, before any gene tree is computed using SplitsTree
- Plan:
    1. Walk through Neighbor-Net on one alignment together
    2. Each participant loads 2-3 more alignments from different categories and compares
    3. Discuss what the network shapes reveal

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Neighbor-Net on our alignments

- We can use SplitsTree> Neighbor-Net to compute split networks for our 15 alignments
- Question: do they all look the same? Where are the parallelograms?

Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
Autosomal (2R, 3R)	5	(arabiensis, (gambiae, coluzzii)) - introgressed
2La inversion (2L)	2	2La-specific topologies
3La inversion (3L)	2	(merus, quadriannulatus) - unexpected introgression

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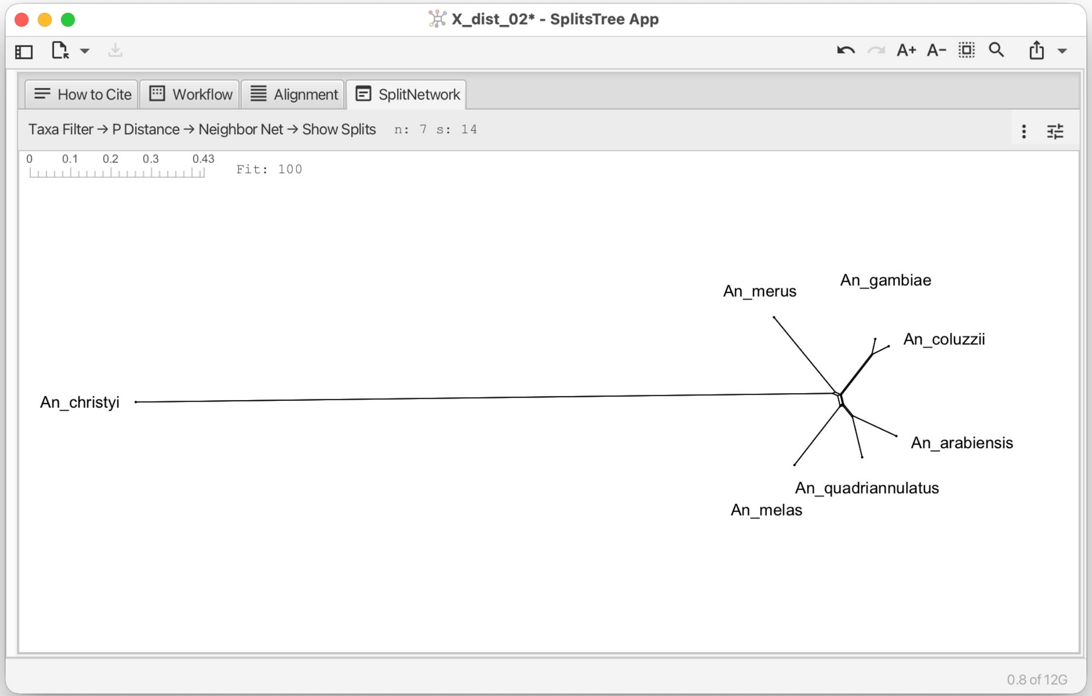
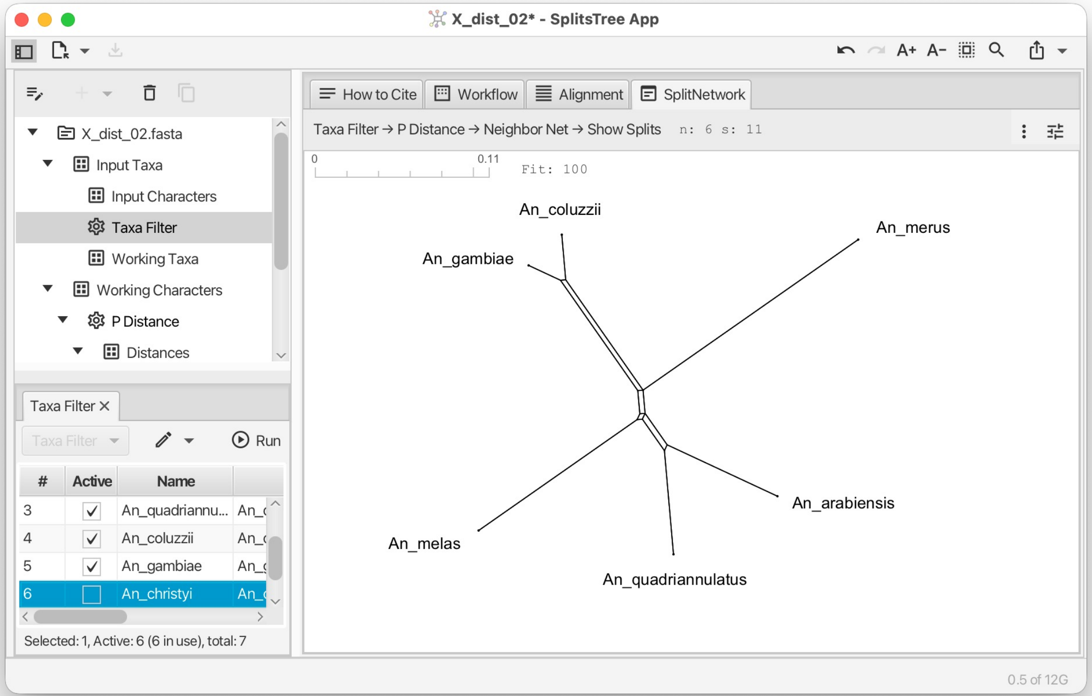
Let's do one together: X_dist_02

- Open file `data/alignments/X_dist_02.fasta` in SplitsTree
- This will run p-distances and Neighbor-Net



Let's do one together: X_dist_02

- Open file `data/alignments/X_dist_02.fasta` in SplitsTree
- Use the Taxa Filter item (side bar) to deactivate the outgroup:



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Open other alignments and explore

- Repeat the same workflow on alignments from different categories:

Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
Autosomal (2R, 3R)	5	(arabiensis, (gambiae, coluzzii)) - introgressed
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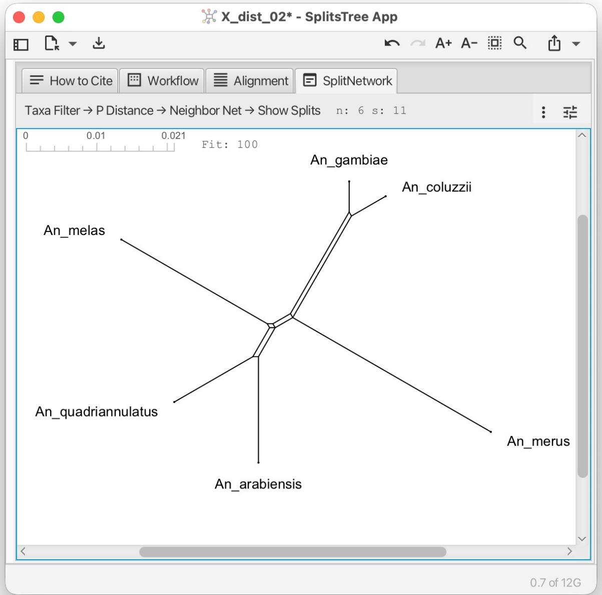
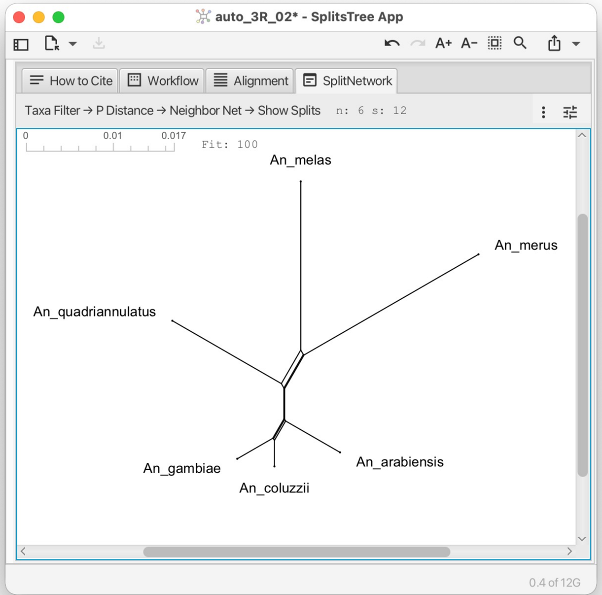
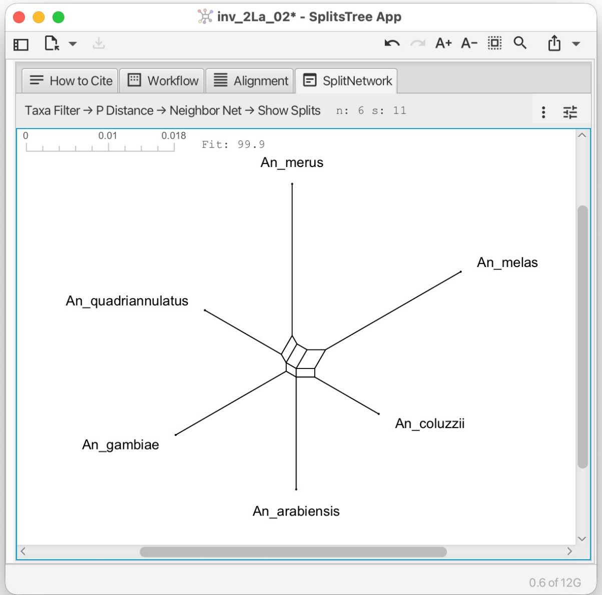
Open other alignments and explore







- X_dist_02
- “species tree”

- auto_3R_02
- “introgressed”

- Inv_2La_02
- “2La-specific topologies”

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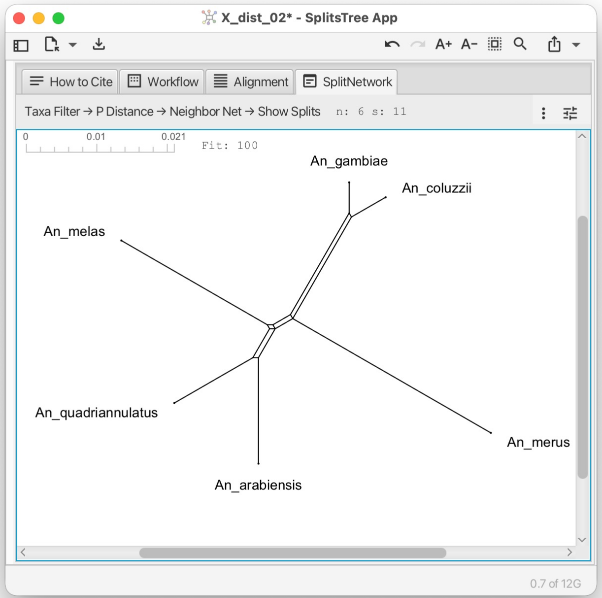
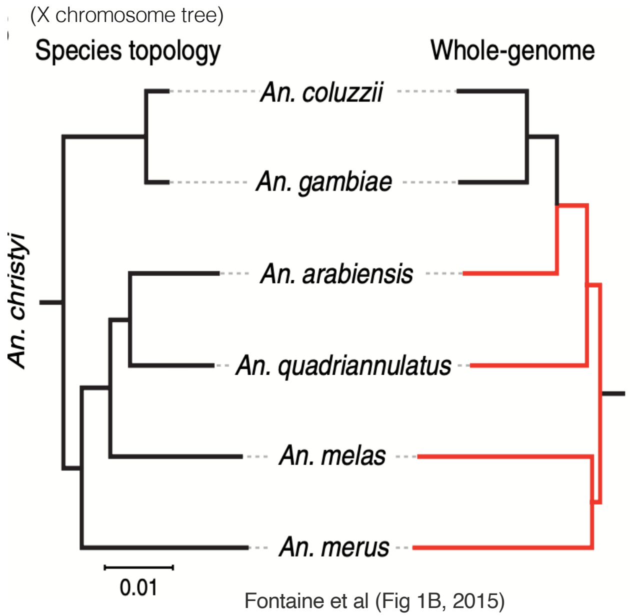
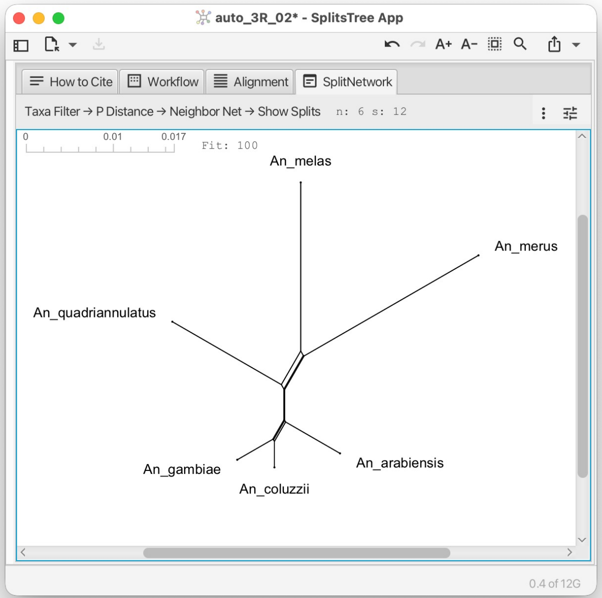
Compare with trees in Fontaine et al (2015)



- X_dist_02
- “species tree”

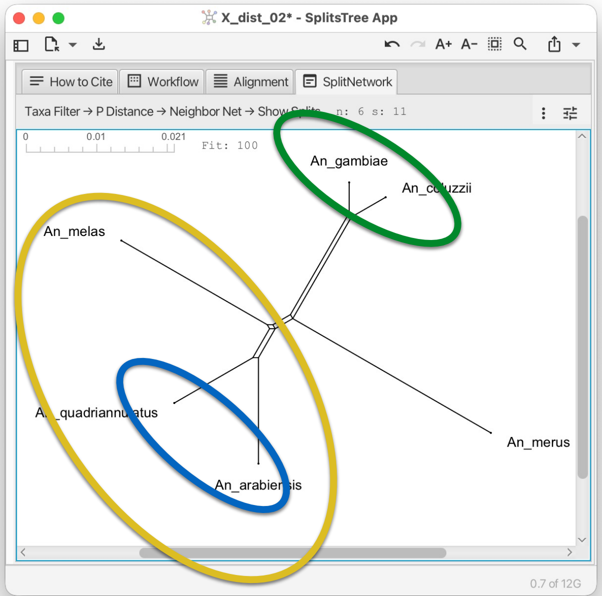


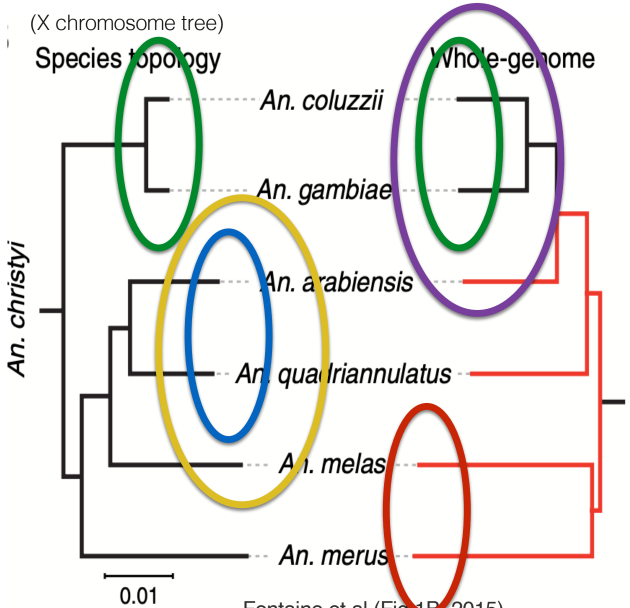
Fontaine et al (Fig 1B, 2015)



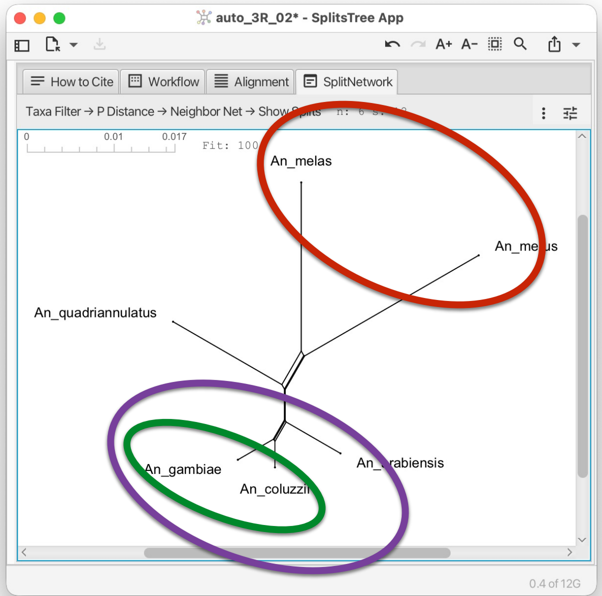
- auto_3R_02
- “introgressed”

Compare with trees in Fontaine et al (2015)





Fontaine et al (Fig 1B, 2015)



- X_dist_02
- “species tree”

- auto_3R_02
- “introgressed”

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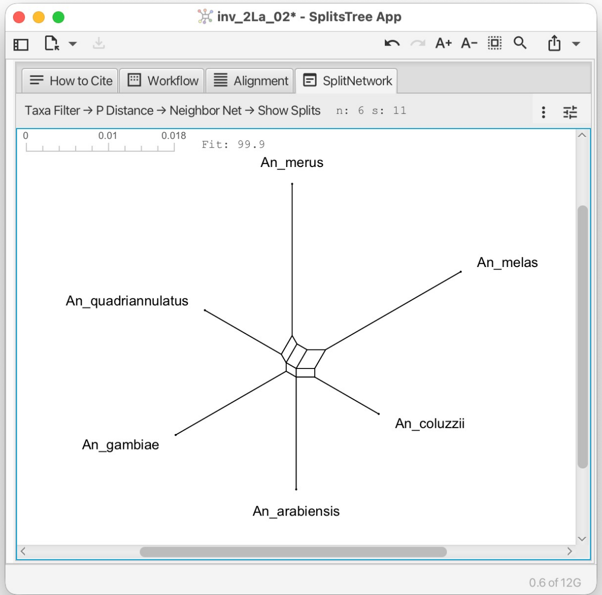
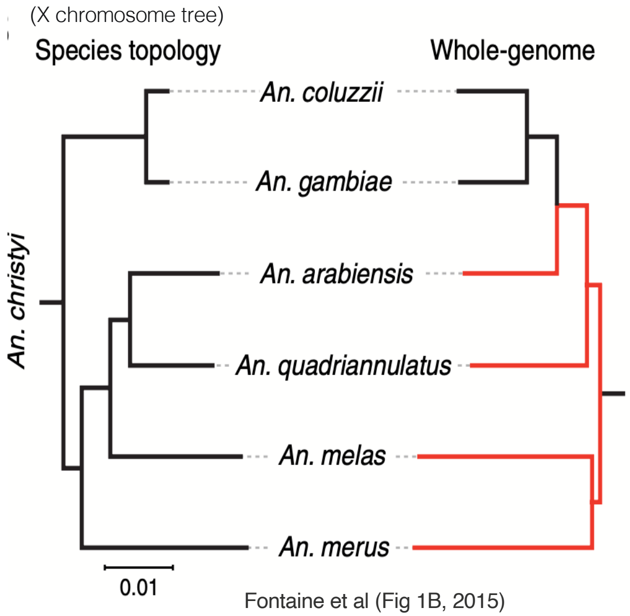
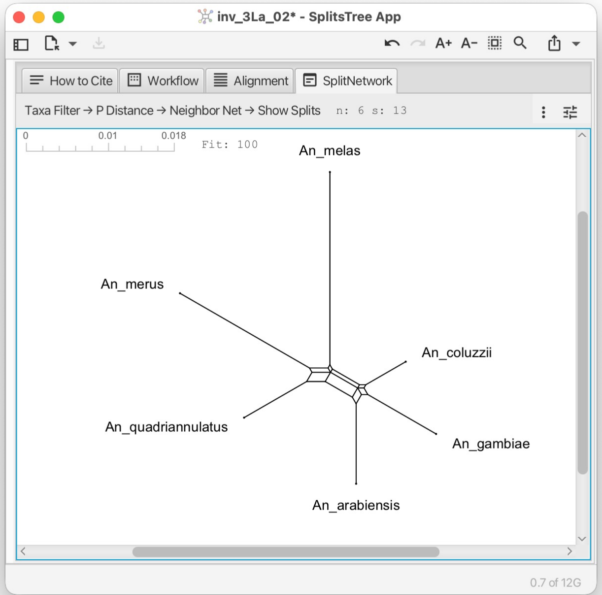
Compare with trees in Fontaine et al (2015)





Fontaine et al (Fig 1B, 2015)



- Inv_2La_02
- "2La-specific topologies"

- Inv_3La_02
- "unexpected introgression"

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What did you see?

- Across the locus categories, you should have seen:
  - X-distal alignments: smaller parallelograms; closer to tree-like
  - Autosomal alignments: more parallelograms involving `arabiensis` and the (gambiae, coluzzii) clade
  - 2La inversion alignments: distinctive boxes reflecting the trans-specific polymorphism
  - 3La inversion alignments: parallelograms grouping `merus` with `quadriannulatus`
- The big-picture observation: the conflict pattern we saw in gene trees from IQ-TREE is also visible directly in alignment data

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And what's next?

PhyloSketch (20 min):

- Neighbor-Net produces an implicit network - it indicates conflict
- For our next step, we will build an explicit network by hand, drawing reticulation nodes representing specific hypothesized evolutionary events
- This is the "phylogenetic sketching" workflow

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Hands-on: sketch a hypothesis network in PhyloSketch

- Time budget: 20 minutes
- Goal: build, by hand, a phylogenetic network that captures your best hypothesis about Anopheles evolution given everything we've seen
- Tool: PhyloSketch (Huson, 2025)
- The pedagogical idea:
  - You have observed patterns in gene trees and Neighbor-Nets
  - Translate those patterns into a biological hypothesis: what actually happened in this group's evolutionary history?
  - Draw your hypothesis as an explicit network with reticulation nodes

PhyloSketch in 90 seconds

- Launch PhyloSketch
- Drawing leaves and branches: click to place taxa, drag to connect them
- Building a tree: connect leaves through internal nodes
- Adding a reticulation: convert an internal node to a reticulation node, then connect a *second* parent edge to it
- Editing labels: double-click to rename
- Saving: File -> Save As -> .psketch file

Build your hypothesis network

Step 1: Start from the species tree we believe is correct

From X-distal loci, the species tree is:

```
((((arabiensis,quadriannulatus),
(gambiae, coluzzii)),melas),merus);
```

with `christyi` as outgroup. Sketch this in PhyloSketch first.

Step 2: Add reticulation nodes for hypothesized introgression

Events to consider:

Event	Evidence in our data
Introgression from (gambiae, coluzzii) into arabiensis	Autosomal gene trees show arabiensis grouped with (gambiae, coluzzii) instead of quadriannulatus
Introgression between merus and quadriannulatus	3La inversion gene trees show (merus, quadriannulatus) sister
Bidirectional gene flow involving the 2La inversion	2La inversion gene trees show distinctive non-species-tree topologies

Build your hypothesis network

Step 3: Decide which events to include

- A minimal network has 1 reticulation
- A richer network might have 2 or 3
- Trade-off: every reticulation must be justified by evidence

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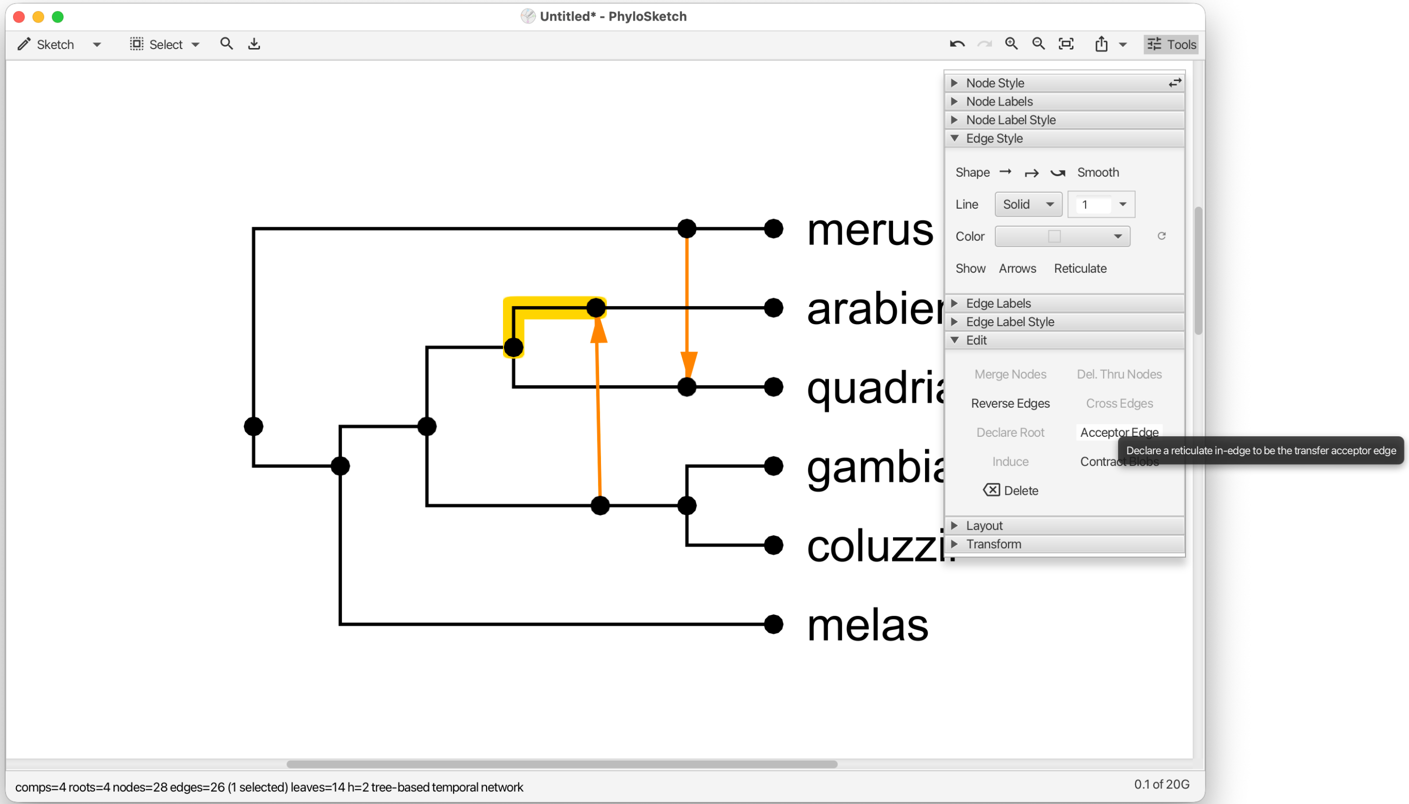
Build your hypothesis network



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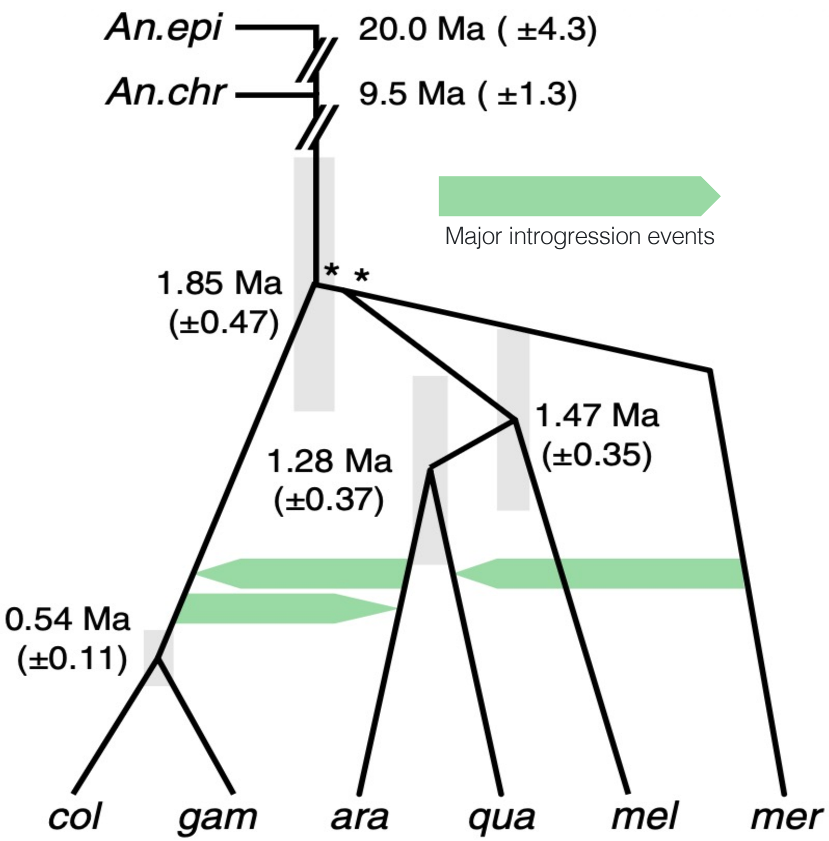
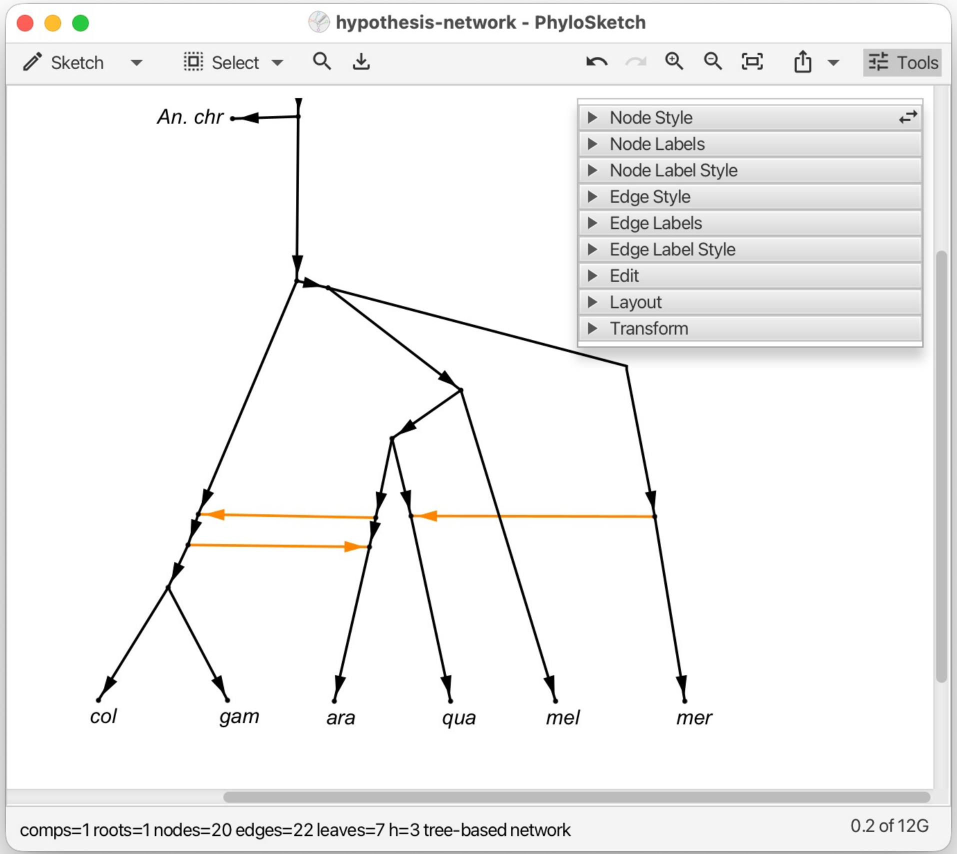
Answer provided in paper (do not peek too early)



Fontaine et al (Fig 1C, 2015)



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PhyloGuide Q4

PhyloGuide
PHYLOGENETICS, GROUNDED

Q: We want to infer an explicit reticulation network from a set of gene trees, with the minimum number of reticulations needed to display all the trees. What method should we use?

A: Use a minimum-hybridization network method, which finds the smallest number of reticulation events explaining all gene trees. Check ILS first because it can mimic reticulation.

Hands-on PhyloCompare: from gene trees to a network

- Time budget: 25 minutes
- Goal: compute an explicit reticulation network from the 15 gene trees you produced in Part I, then interactively explore how different subsets of trees give different networks
- The big question for this segment: does the network agree with what you sketched in PhyloSketch?

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Hands-on PhyloCompare: from gene trees to a network

PhyloCompare

- Novel comparison tool for trees (under development)
- It uses the PhyloFusion algorithm to computed a minimum-reticulate network that contains all input trees (Zhang, Cetinkaya, Huson, 2026)
- We will use the tool for network inference
- Each reticulation in the output is a hypothesized event - matching a hybridization, introgression, or other gene-flow event

Hands-on PhyloCompare: from gene trees to a network

1. Launch PhyloCompare
2. File -> Open -> select `tutorial_loci.nex` (the NEXUS tree set we created after IQ-TREE; it contains all 15 gene trees with their locus IDs)
3. PhyloCompare should list all 15 trees in the tree-set panel; by default, the first trees are combined into a network
4. Explore different sets of trees to run comparisons on

Hands-on PhyloCompare: from gene trees to a network

The point of the tool is interactive comparison. Try the following subsets and see how the network changes:

- All 15 trees together.
- X-distal loci only (5 trees).
- Autosomal loci only (5 trees).
- 2La inversion loci only (2 trees).
- 3La inversion loci only (2 trees).
- Mixed subsets, e.g. X-distal + autosomal, X-distal + 3La.

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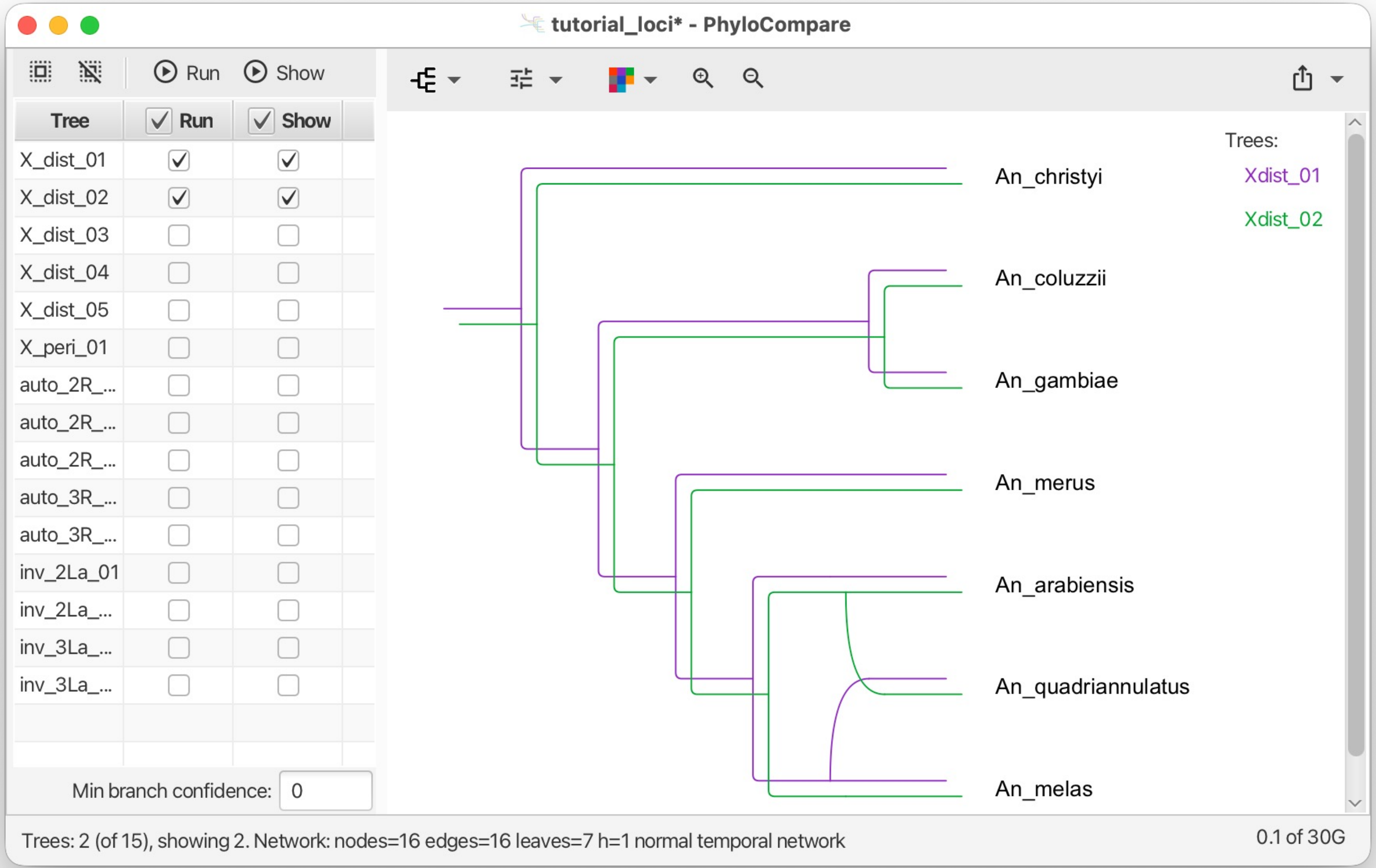
Hands-on PhyloCompare: from gene trees to a network



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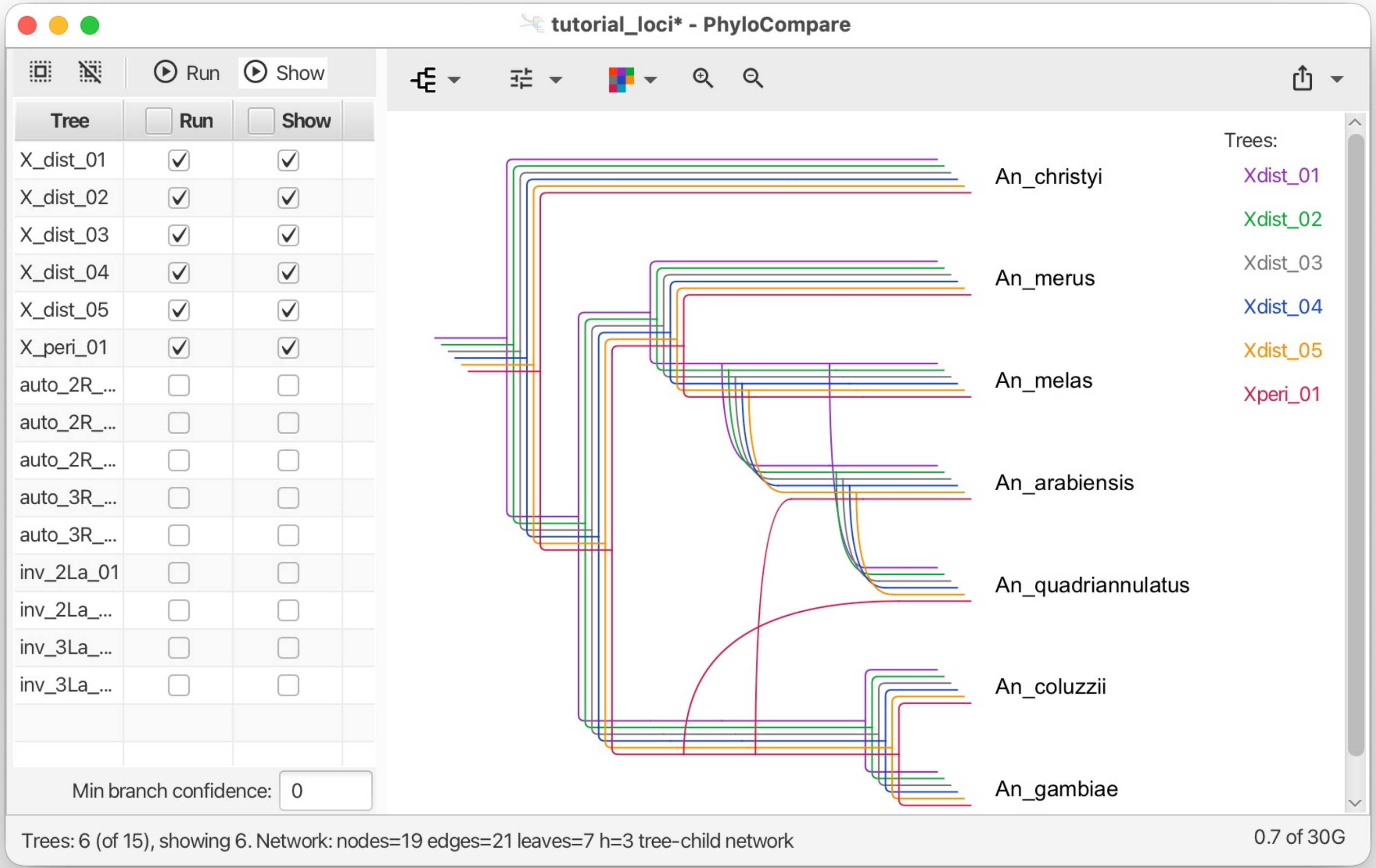
Hands-on PhyloCompare: from gene trees to a network



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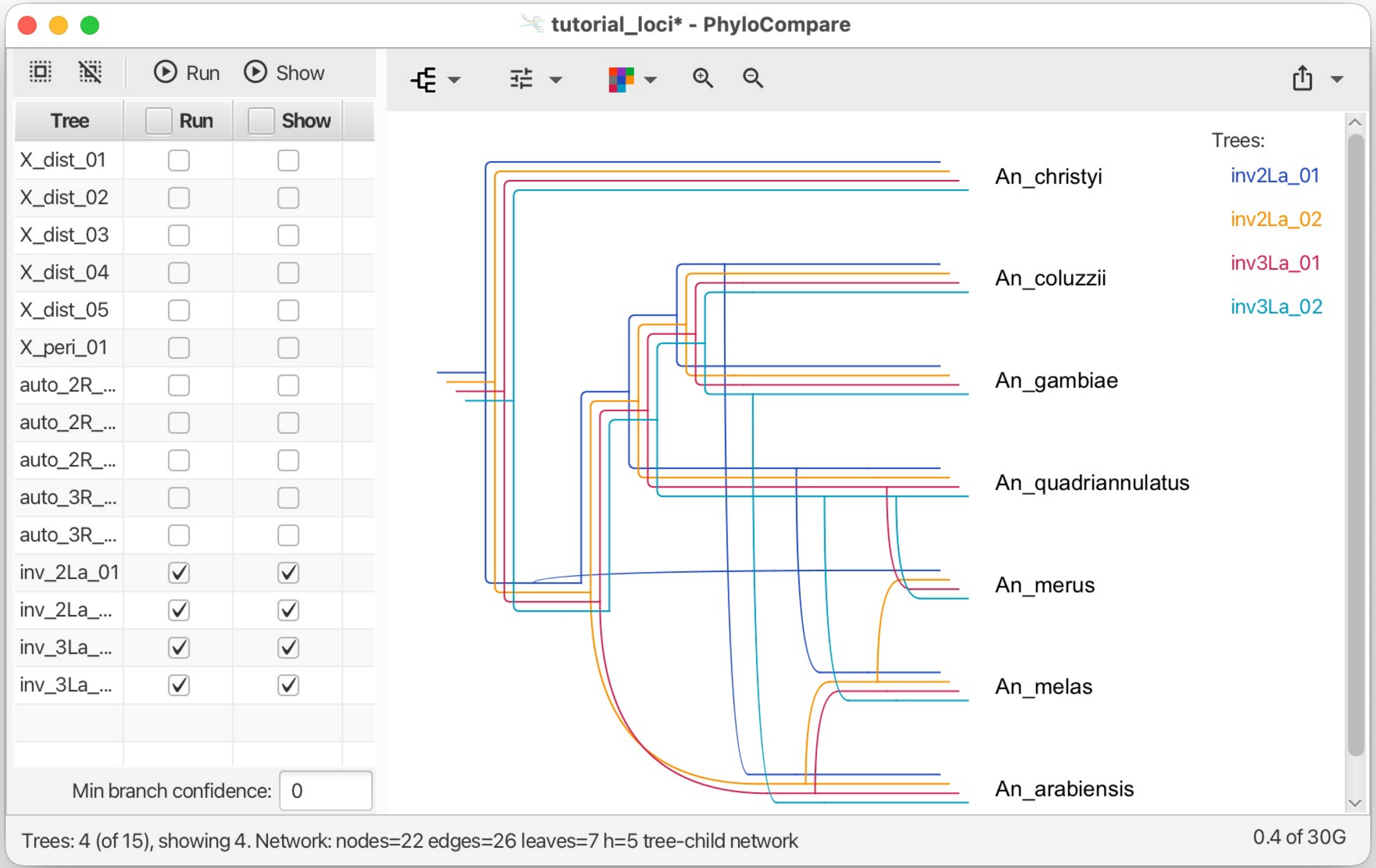
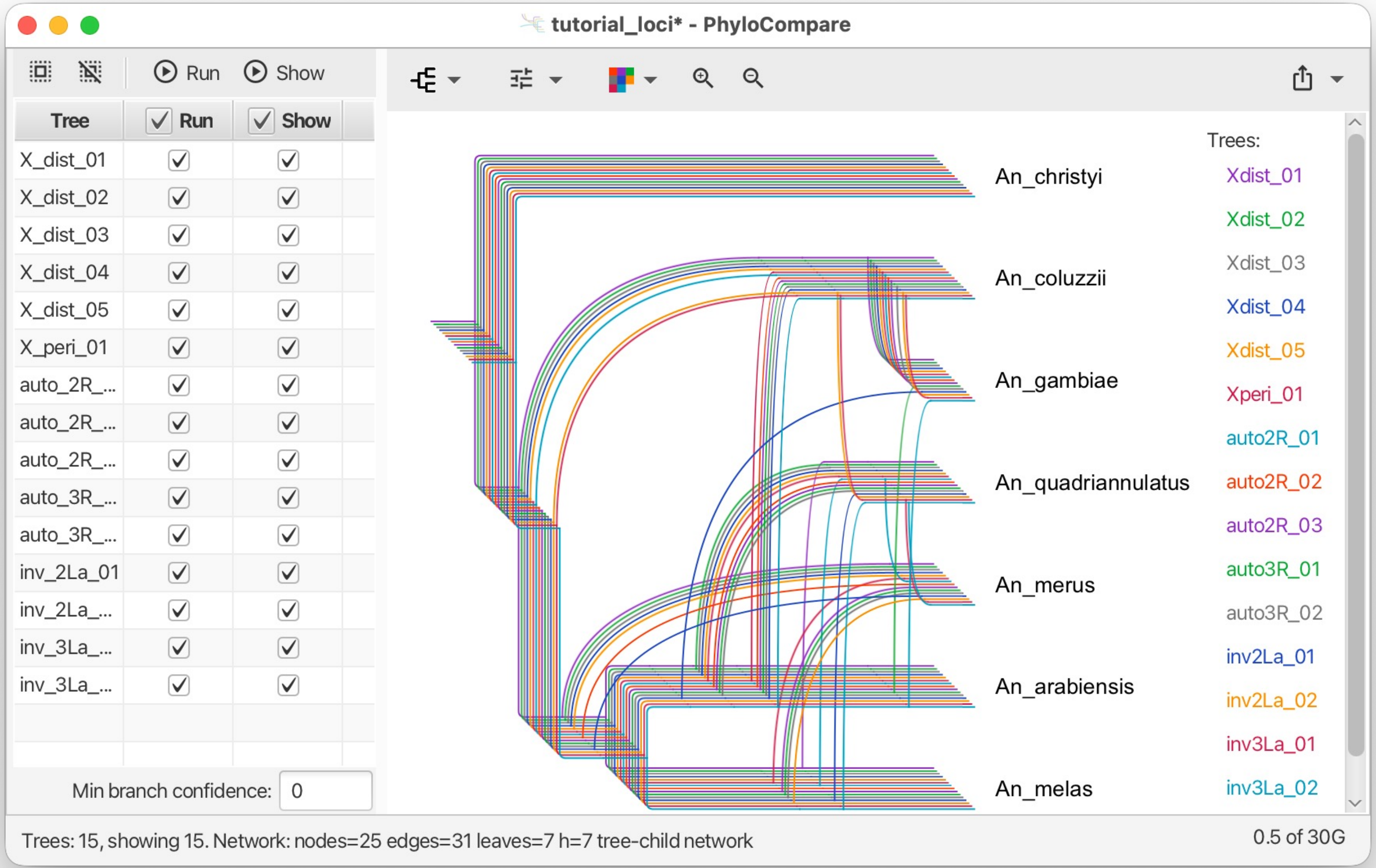
Hands-on PhyloCompare: from gene trees to a network



Hands-on PhyloCompare: from gene trees to a network



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How does PhyloFusion's network compare to what you sketched?

- Open your saved sketch from PhyloSketch alongside the PhyloCompare network
- Compare:
  - Same number of reticulations?
  - Same reticulation events (same pairs of lineages involved)?
  - Same direction of gene flow (if your sketch indicated direction)?

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PhyloGuide Q5

PhyloGuide
PHYLOGENETICS, GROUNDED

Q: We have a phylogenetic network for the Anopheles gambiae species complex with reticulations between An. arabiensis and the (An. gambiae, An. coluzzii) clade, and between An. merus and An. quadriannulatus. What does this tell us biologically?

A: The reticulations suggest non-tree-like evolution, possibly due to historical gene flow/hybridization or ILS. Different loci may reflect different evolutionary histories across the genome.

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Agenda

1. Tutorial overview and dataset
2. Setup verification
3. Multi-locus phylogenomics lecture
4. IQ-TREE 3 hands-on
5. BEAST X + SplitsTree posterior visualization
6. ASTRAL hands-on ☕
7. Networks lecture
8. SplitsTree Neighbor-Net hands-on
9. PhyloSketch hands-on
10. PhyloCompare hands-on
11. Biological interpretation + wrap-up

Back to the biology: what did this tell us about malaria

- The species branching order: X chromosome topology is correct; *An. arabiensis* and *An. quadriannulatus* are sister taxa, NOT the three vector species clustering
- Pervasive autosomal introgression: especially between *An. arabiensis* and the (*An. gambiae, An. coluzzii*) ancestor
- Additional introgression between *An. merus* and *An. quadriannulatus* in the 3La inversion region
- The implication for malaria control:

  If introgression moves vector-capability traits, then control strategies that target one species may be partially neutralized by introgression from another.

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Take-aways from the tutorial

1. Gene trees disagree systematically across the genome when introgression is pervasive. This is data, not noise.
2. Tree-summary methods can be confidently wrong when the underlying biology is reticulate. ASTRAL got the wrong species tree on our data – because it does not model introgression.
3. Networks can make the conflict visible in three ways:
   - alignment-level (Neighbor-Net),
   - within-locus uncertainty (consensus from posterior), and
   - across-locus reticulation (PhyloCompare).
4. Hypothesis sketching before automated inference clarifies what you're looking for and makes the algorithm's output interpretable.
5. AI assistance (PhyloGuide) is useful for orientation and recommendations but should not replace expert judgment.

Thank you - and please give us feedback

- Thanks to:
  - All of you for spending the morning with us
  - The Anopheles genomics consortium and Fontaine et al. for the dataset that made this possible
  - The tool developers whose work underlies the tutorial (Minh lab, Suchard lab, Drummond lab, Rambaut lab, Mirarab lab, and our own group)
  - ISCB for the tutorial program
  - University of Tübingen for support

Share Your Feedback

Scan the QR code to let us know your thoughts on this tutorial.
Thank you!

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